

TL08xx FET-Input Operational Amplifiers

1 Features

- High slew rate: 20 V/μs (TL08xH, typ)
- Low offset voltage: 1 mV (TL08xH, typ)
- Low offset voltage drift: 2 μV/°C
- Low power consumption: 940 μA/ch (TL08xH, typ)
- Wide common-mode and differential voltage ranges
 - Common-mode input voltage range includes V_{CC+}
- Low input bias and offset currents
- Low noise:
 $V_n = 18 \text{ nV}/\sqrt{\text{Hz}}$ (typ) at $f = 1 \text{ kHz}$
- Output short-circuit protection
- Low total harmonic distortion: 0.003% (typ)
- Wide supply voltage:
 $\pm 2.25 \text{ V}$ to $\pm 20 \text{ V}$, 4.5 V to 40 V

2 Applications

- [Solar energy: string and central inverter](#)
- [Motor drives: AC and servo drive control and power stage modules](#)
- [Single phase online UPS](#)
- [Three phase UPS](#)
- [Pro audio mixers](#)
- [Battery test equipment](#)

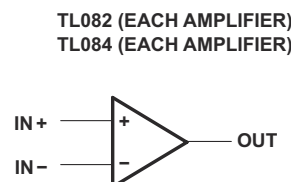
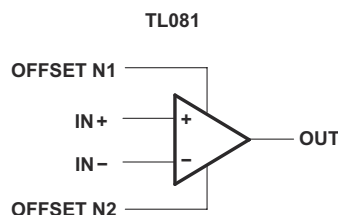
3 Description

The TL08xH (TL081H, TL082H, and TL084H) family of devices are the next-generation versions of the industry-standard TL08x (TL081, TL082, and TL084) devices. These devices provide outstanding value for cost-sensitive applications, with features including low offset (1 mV, typical), high slew rate (20 V/μs), and common-mode input to the positive supply. High ESD (1.5 kV, HBM), integrated EMI and RF filters, and operation across the full -40°C to 125°C enable the TL08xH devices to be used in the most rugged and demanding applications.

Device Information

PART NUMBER ⁽¹⁾	PACKAGE	BODY SIZE (NOM)
TL081x	PDIP (8)	9.59 mm × 6.35 mm
	SC70 (5)	2.00 mm × 1.25 mm
	SO (8)	6.20 mm × 5.30 mm
	SOIC (8)	4.90 mm × 3.90 mm
	SOT-23 (5)	1.60 mm × 1.20 mm
TL082x	PDIP (8)	9.59 mm × 6.35 mm
	SO (8)	6.20 mm × 5.30 mm
	SOIC (8)	4.90 mm × 3.90 mm
	SOT-23 (8)	2.90 mm × 1.60 mm
	TSSOP (8)	4.40 mm × 3.00 mm
TL082M	CDIP (8)	9.59 mm × 6.67 mm
	LCCC (20)	8.89 mm × 8.89 mm
TL084x	PDIP (14)	19.30 mm × 6.35 mm
	SO (14)	10.30 mm × 5.30 mm
	SOIC (14)	8.65 mm × 3.91 mm
	SOT-23 (14)	4.20 mm × 2.00 mm
	TSSOP (14)	5.00 mm × 4.40 mm
TL084M	CDIP (14)	19.56 mm × 6.92 mm
	LCCC (20)	8.89 mm × 8.89 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Logic Symbols



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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision L (July 2021) to Revision M (December 2021)	Page
• Corrected DCK pinout diagram and table in <i>Pin Configurations and Functions</i> section.....	4

Changes from Revision K (June 2021) to Revision L (July 2021)	Page
• Deleted preview note from TL081H SOIC (8), SOT-23 (5), and SC70 (5) packages throughout the data sheet	1

Changes from Revision J (November 2020) to Revision K (June 2021)	Page
• Deleted VSSOP (8) package references throughout data sheet.....	1
• Deleted preview note from TL082H SOIC (8), SOT-23 (8), and TSSOP (8) packages throughout the data sheet.....	1
• Added DBV, DCK, and D packages to TL081H in <i>Pin Configuration and Functions</i> section.....	4
• Added ESD information for TL082H.....	10
• Added D, DCK, and DBV package thermal information in Thermal Information for Single Channel: TL081H section.....	11
• Added D, DDF, and PW package thermal information in Thermal Information for Dual Channel: TL082H section	11
• Added I _B and I _{OS} specification for single channel DCK and DBV package.....	13
• Added I _Q spec for TL081H and TL082H.....	13
• Removed <i>Related Links</i> section from <i>Device and Documentation Support</i> section.....	36

Changes from Revision I (May 2015) to Revision J (November 2020) Page

• Updated the numbering format for tables, figures, and cross-references throughout the document.....	1
• Added TL08xH devices throughout the data sheet.....	1
• Added features for TL08xH to the <i>Features</i> section.....	1
• Added link to applications in the <i>Applications</i> section.....	1
• Added TL08xH in the <i>Description</i> section.....	1
• Added TL08xH in the <i>Device Information</i> table.....	1
• Updated pinout diagrams and pinout tables in <i>Pin Configurations and Functions</i> section	4
• Added TSSOP, VSSOP and DDF packages to TL082x in <i>Pin Configuration and Functions</i> section.....	4
• Added DYY package to TL084x in <i>Pin Configuration and Functions</i> section.....	4
• Added <i>Typical Characteristics: TL08xH</i> section in <i>Specifications</i> section.....	18
• Removed Table of Graphs in <i>Typical Characteristics: All Other Devices</i> section.....	25
• Removed references to obsolete documentation.....	35

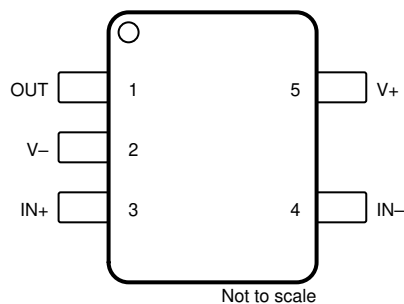
Changes from Revision H (January 2014) to Revision I (May 2015) Page

• Added <i>Applications</i> section, <i>Device Information</i> table, Pin Functions table, Thermal Information table, <i>Feature Description</i> section, <i>Device Functional Modes</i> section, <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, ESD information, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section.....	1
• Added <i>Applications</i>	1
• Moved <i>Typical Characteristics</i> into <i>Specifications</i> section.	25

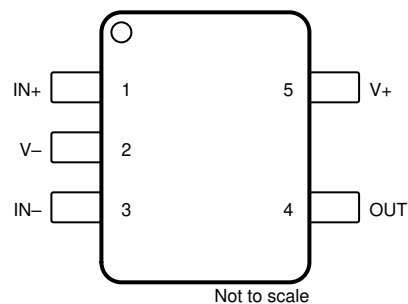
Changes from Revision G (September 2004) to Revision H (January 2014) Page

• Deleted <i>Ordering Information</i> table.....	1
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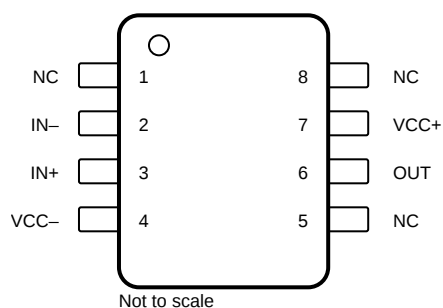
5 Pin Configuration and Functions



**Figure 5-1. TL081H DBV Package
5-Pin SOT-23
(Top View)**



**Figure 5-2. TL081H DCK Package
5-Pin SC70
(Top View)**

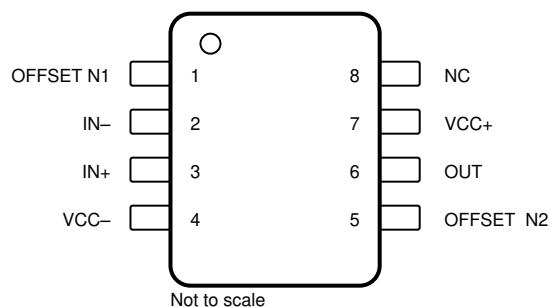


NC- no internal connection

**Figure 5-3. TL081H D Package
8-Pin SOIC
(Top View)**

Table 5-1. Pin Functions: TL081H

NAME	PIN			I/O	DESCRIPTION
	DBV	DCK	D		
IN–	4	3	2	I	Inverting input
IN+	3	1	3	I	Noninverting input
NC	—	—	8	—	Do not connect
NC	—	—	1	—	Do not connect
NC	—	—	5	—	Do not connect
OUT	1	4	6	O	Output
VCC–	2	2	4	—	Power supply
VCC+	5	5	7	—	Power supply

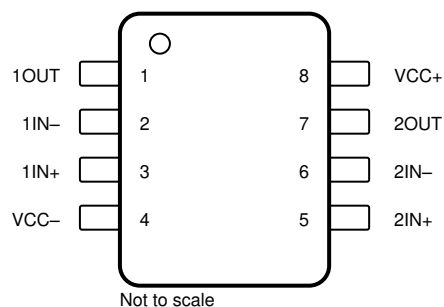


NC- no internal connection

**Figure 5-4. TL081x D, P, and PS Package
8-Pin SOIC, PDIP, and SO
(Top View)**

Table 5-2. Pin Functions: TL081x

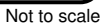
PIN		I/O	DESCRIPTION
NAME	NO.		
IN–	2	I	Inverting input
IN+	3	I	Noninverting input
NC	8	—	Do not connect
OFFSET N1	1	—	Input offset adjustment
OFFSET N2	5	—	Input offset adjustment
OUT	6	O	Output
VCC–	4	—	Power supply
VCC+	7	—	Power supply



**Figure 5-5. TL082x D, DDF, DGK, JG, P, PS, and PW Package
 8-Pin SOIC, SOT-23 (8), VSSOP, CDIP, PDIP, SO, and TSSOP
 (Top View)**

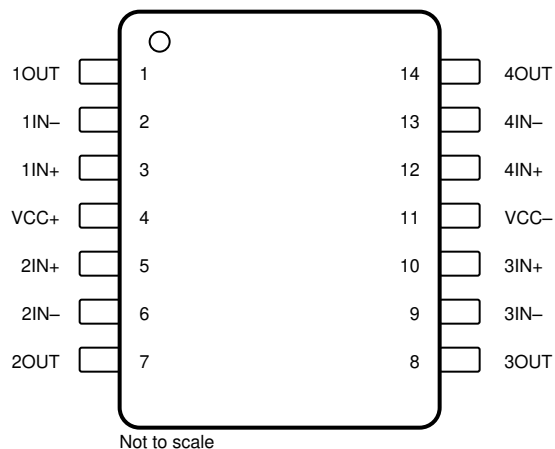
Table 5-3. Pin Functions: TL082x

PIN		I/O	DESCRIPTION
NAME	NO.		
1IN-	2	I	Inverting input
1IN+	3	I	Noninverting input
1OUT	1	O	Output
2IN-	6	I	Inverting input
2IN+	5	I	Noninverting input
2OUT	7	O	Output
VCC-	4	—	Power supply
VCC+	8	—	Power supply



**Figure 5-6. TL082 FK Package
20-Pin LCCC
(Top View)**

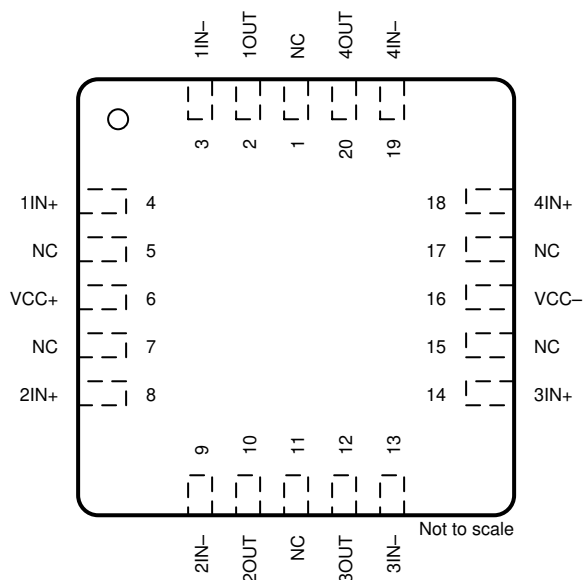
PIN		I/O	DESCRIPTION
NAME	NO.		
1IN−	5	I	Inverting input
1IN+	7	I	Noninverting input
1OUT	2	O	Output
2IN−	15	I	Inverting input
2IN+	12	I	Noninverting input
2OUT	17	O	Output
NC	1, 3, 4, 6, 8, 9, 11, 13, 14, 16, 18, 19	—	Do not connect
VCC−	10	—	Power supply
VCC+	20	—	Power supply



**Figure 5-7. TL084x D, N, NS, PW, J, and DYY Package
 14-Pin SOIC, PDIP, SO, TSSOP, CDIP, and SOT-23 (14)
 (Top View)**

Table 5-5. Pin Functions: TL084x

PIN		I/O	DESCRIPTION
NAME	NO.		
1IN-	2	I	Inverting input
1IN+	3	I	Noninverting input
1OUT	1	O	Output
2IN-	6	I	Inverting input
2IN+	5	I	Noninverting input
2OUT	7	O	Output
3IN-	9	I	Inverting input
3IN+	10	I	Noninverting input
3OUT	8	O	Output
4IN-	13	I	Inverting input
4IN+	12	I	Noninverting input
4OUT	14	O	Output
V _{CC-}	11	—	Power supply
V _{CC+}	4	—	Power supply



NC- no internal connection

**Figure 5-8. TL084 FK Package
20-Pin LCCC
(Top View)**

Table 5-6. Pin Functions: TL084x

PIN		I/O	DESCRIPTION
NAME	NO.		
1IN-	3	I	Inverting input
1IN+	4	I	Noninverting input
1OUT	2	O	Output
2IN-	9	I	Inverting input
2IN+	8	I	Noninverting input
2OUT	10	O	Output
3IN-	13	I	Inverting input
3IN+	14	I	Noninverting input
3OUT	12	O	Output
4IN-	19	I	Inverting input
4IN+	18	I	Noninverting input
4OUT	20	O	Output
NC	1, 5, 7, 11, 15, 17	—	Do not connect
VCC-	16	—	Power supply
VCC+	6	—	Power supply

6 Specifications

6.1 Absolute Maximum Ratings: TL08xH

over operating ambient temperature range (unless otherwise noted) ⁽¹⁾

		MIN	MAX	UNIT
Supply voltage, $V_S = (V_{CC+}) - (V_{CC-})$		0	42	V
Signal input pins	Common-mode voltage ⁽³⁾	$(V_{CC-}) - 0.5$	$(V_{CC+}) + 0.5$	V
	Differential voltage ⁽³⁾		$V_S + 0.2$	V
	Current ⁽³⁾	-10	10	mA
Output short-circuit ⁽²⁾		Continuous		
Operating ambient temperature, T_A		-55	150	°C
Junction temperature, T_J			150	°C
Storage temperature, T_{stg}		-65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) Short-circuit to ground, one amplifier per package.
- (3) Input pins are diode-clamped to the power-supply rails. Input signals that may swing more than 0.5 V beyond the supply rails must be current limited to 10 mA or less.

6.2 Absolute Maximum Ratings: All Other Devices

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

		MIN	MAX	UNIT
$V_{CC+} - V_{CC-}$	Supply voltage ⁽²⁾	-18	18	V
V_{ID}	Differential input voltage ⁽³⁾	-30	+30	V
V_I	Input voltage ^{(2) (4)}	-15	+15	V
	Duration of output short circuit ⁽⁵⁾	Unlimited		
	Continuous total power dissipation	See Section 6.15		
T_{stg}	Storage temperature	-65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) All voltage values, except differential voltages, are with respect to the midpoint between V_{CC+} and V_{CC-} .
- (3) Differential voltages are at $IN+$, with respect to $IN-$.
- (4) The magnitude of the input voltage must never exceed the magnitude of the supply voltage or 15 V, whichever is less.
- (5) The output may be shorted to ground or to either supply. Temperature and/or supply voltages must be limited to ensure that the dissipation rating is not exceeded.

6.3 ESD Ratings: TL08xH

			VALUE	UNIT
TL084H				
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±1500	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000	
TL082H and TL081H				
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.4 ESD Ratings: All Other Devices

		VALUE	UNIT
V_{ESD} Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±1000	V
	Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1500	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.5 Recommended Operating Conditions: TL08xH

over operating ambient temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V_S	Supply voltage, $(V_{CC+}) - (V_{CC-})$	4.5	40	V
V_I	Input voltage range	$(V_{CC-}) + 2$	$(V_{CC+}) + 0.1$	V
T_A	Specified temperature	–40	125	°C

6.6 Recommended Operating Conditions: All Other Devices

over operating free-air temperature range (unless otherwise noted)

			MIN	MAX	UNIT
V _{CC+}	Supply voltage		5	15	V
V _{CC−}	Supply voltage		−5	−15	V
V _{CM}	Common-mode voltage		V _{CC−} + 4	V _{CC+} − 4	V
T _A	Ambient temperature	TL08xM	−55	125	°C
		TL08xQ	−40	125	
		TL08xI	−40	85	
		TL08xC	0	70	

6.7 Thermal Information for Single Channel: TL081H

THERMAL METRIC ⁽¹⁾		TL081H			UNIT
		D (SOIC)	DCK (SC70)	DBV (SOT-23)	
		8 PINS	5 PINS	5 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	158.8	217.5	212.2	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	98.6	113.1	111.1	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	102.3	63.8	79.4	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	45.8	34.8	51.8	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	101.5	63.5	79.0	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.8 Thermal Information for Dual Channel: TL082H

THERMAL METRIC ⁽¹⁾		TL082H			UNIT
		D (SOIC)	DDF (SOT-23)	PW (TSSOP)	
		8 PINS	8 PINS	8 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	147.8	181.5	200.3	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	88.2	112.5	89.4	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	91.4	98.2	131.0	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	36.8	17.2	22.0	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	90.6	97.6	129.3	°C/W

6.8 Thermal Information for Dual Channel: TL082H (continued)

THERMAL METRIC ⁽¹⁾		TL082H			UNIT
		D (SOIC)	DDF (SOT-23)	PW (TSSOP)	
		8 PINS	8 PINS	8 PINS	
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.9 Thermal Information for Quad Channel: TL084H

THERMAL METRIC ⁽¹⁾		TL084H			UNIT
		D (SOIC)	DYY ⁽²⁾ (SOT-23)	PW (TSSOP)	
		14 PINS	14 PINS	14 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	114.2	TBD	134.4	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	70.3	TBD	62.6	°C/W
R _{θJB}	Junction-to-board thermal resistance	70.2	TBD	77.6	°C/W
ψ _{JT}	Junction-to-top characterization parameter	28.8	TBD	13.0	°C/W
ψ _{JB}	Junction-to-board characterization parameter	69.8	TBD	77.0	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	TBD	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).
 (2) This package option is preview for TL084H.

6.10 Thermal Information: All Other Devices

THERMAL METRIC ⁽¹⁾		TL08xxx											UNIT
		D (SOIC)		FK (LCCC)	J (CDIP)		N (PDIP)		NS (SO)		PW (TSSOP)		
		8 PIN	14 PIN	20 PIN	8 PIN	14 PIN	8 PIN	14 PIN	8 PIN	14 PIN	8 PIN	14 PIN	
R _{θJA}	Junction-to-ambient thermal resistance	97	86				85	80	95	76	150	113	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance			5.61	15.05	14.5							

- (1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, [SPRA953](#).

6.11 Electrical Characteristics: TL08xH

For $V_S = (V_{CC+}) - (V_{CC-}) = 4.5 \text{ V to } 40 \text{ V}$ ($\pm 2.25 \text{ V to } \pm 20 \text{ V}$) at $T_A = 25^\circ\text{C}$, $R_L = 10 \text{ k}\Omega$ connected to $V_S / 2$, $V_{CM} = V_S / 2$, and $V_{OUT} = V_S / 2$, unless otherwise noted.

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
OFFSET VOLTAGE							
V _{OS}	Input offset voltage			±1	±4	mV	
			T _A = −40°C to 125°C		±5		
dV _{OS} /dT	Input offset voltage drift		T _A = −40°C to 125°C	±2		μV/°C	
PSRR	Input offset voltage versus power supply	V _S = 5 V to 40 V, V _{CM} = V _S / 2	T _A = −40°C to 125°C	±1	±10	μV/V	
	Channel separation	f = 0 Hz		10		μV/V	
INPUT BIAS CURRENT							
I _B	Input bias current			±1	±120	pA	
			DCK and DBV packages	±1	±300	pA	
			T _A = −40°C to 125°C ⁽¹⁾		±5	nA	
I _{OS}	Input offset current			±0.5	±120	pA	
			DCK and DBV packages	±0.5	±250	pA	
			T _A = −40°C to 125°C ⁽¹⁾		±5	nA	
NOISE							
E _N	Input voltage noise	f = 0.1 Hz to 10 Hz		9.2		μV _{PP}	
				1.4		μV _{RMS}	
e _N	Input voltage noise density	f = 1 kHz		37		nV/√Hz	
		f = 10 kHz		21			
i _N	Input current noise	f = 1 kHz		80		fA/√Hz	
INPUT VOLTAGE RANGE							
V _{CM}	Common-mode voltage range			(V _{CC−}) + 1.5	(V _{CC+})	V	
CMRR	Common-mode rejection ratio	V _S = 40 V, (V _{CC−}) + 2.5 V < V _{CM} < (V _{CC+}) − 1.5 V		100	105	dB	
CMRR	Common-mode rejection ratio		T _A = −40°C to 125°C	95		dB	
CMRR	Common-mode rejection ratio	V _S = 40 V, (V _{CC−}) + 2.5 V < V _{CM} < (V _{CC+})		90	105	dB	
CMRR	Common-mode rejection ratio		T _A = −40°C to 125°C	80		dB	
INPUT CAPACITANCE							
Z _{ID}	Differential			100 2		MΩ pF	
Z _{ICM}	Common-mode			6 1		TΩ pF	
OPEN-LOOP GAIN							
A _{OL}	Open-loop voltage gain	V _S = 40 V, V _{CM} = V _S / 2, (V _{CC−}) + 0.3 V < V _O < (V _{CC+}) − 0.3 V	T _A = −40°C to 125°C	118	125	dB	
A _{OL}	Open-loop voltage gain	V _S = 40 V, V _{CM} = V _S / 2, R _L = 2 kΩ, (V _{CC−}) + 1.2 V < V _O < (V _{CC+}) − 1.2 V	T _A = −40°C to 125°C	115	120	dB	
FREQUENCY RESPONSE							
GBW	Gain-bandwidth product			5.25		MHz	
SR	Slew rate	V _S = 40 V, G = +1, C _L = 20 pF		20		V/μs	
t _S	Settling time	To 0.1%, V _S = 40 V, V _{STEP} = 10 V , G = +1, CL = 20 pF		0.63		μs	
		To 0.1%, V _S = 40 V, V _{STEP} = 2 V , G = +1, CL = 20 pF		0.56			
		To 0.01%, V _S = 40 V, V _{STEP} = 10 V , G = +1, CL = 20 pF		0.91			
		To 0.01%, V _S = 40 V, V _{STEP} = 2 V , G = +1, CL = 20 pF		0.48			
	Phase margin	G = +1, R _L = 10kΩ, C _L = 20 pF		56		°	
	Overload recovery time	V _{IN} × gain > V _S		300		ns	

6.11 Electrical Characteristics: TL08xH (continued)

For $V_S = (V_{CC+}) - (V_{CC-}) = 4.5 \text{ V to } 40 \text{ V}$ ($\pm 2.25 \text{ V to } \pm 20 \text{ V}$) at $T_A = 25^\circ\text{C}$, $R_L = 10 \text{ k}\Omega$ connected to $V_S / 2$, $V_{CM} = V_S / 2$, and $V_{O\ UT} = V_S / 2$, unless otherwise noted.

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
THD+N	Total harmonic distortion + noise	$V_S = 40\text{ V}$, $V_O = 6\text{ V}_{\text{RMS}}$, $G = +1$, $f = 1\text{ kHz}$		0.00012			%
EMIRR	EMI rejection ratio	$f = 1\text{ GHz}$		53			dB
OUTPUT							
	Voltage output swing from rail	Positive rail headroom	$V_S = 40\text{ V}$, $R_L = 10\text{ k}\Omega$	115	210	mV	
			$V_S = 40\text{ V}$, $R_L = 2\text{ k}\Omega$	520	965		
		Negative rail headroom	$V_S = 40\text{ V}$, $R_L = 10\text{ k}\Omega$	105	215		
			$V_S = 40\text{ V}$, $R_L = 2\text{ k}\Omega$	500	1030		
I_{SC}	Short-circuit current			± 26			mA
C_{LOAD}	Capacitive load drive			300			pF
Z_O	Open-loop output impedance	$f = 1\text{ MHz}$, $I_O = 0\text{ A}$		125			Ω
POWER SUPPLY							
I_Q	Quiescent current per amplifier	$I_O = 0\text{ A}$		937.5	1125	μA	
		$I_O = 0\text{ A}$, (TL081H)		960	1156		
		$I_O = 0\text{ A}$	$T_A = -40^{\circ}\text{C to } 125^{\circ}\text{C}$		1130		
		$I_O = 0\text{ A}$, (TL082H)		1143			
		$I_O = 0\text{ A}$, (TL081H)		1160			
	Turn-On Time	$\text{At } T_A = 25^{\circ}\text{C}$, $V_S = 40\text{ V}$, V_S ramp rate $> 0.3\text{ V}/\mu\text{s}$		60		μs	

(1) Max I_B and I_{OS} data is specified based on characterization results.

6.12 Electrical Characteristics for TL08xC, TL08xxC, and TL08xI

$V_{CC\pm} = \pm 15\text{ V}$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS	T_A ⁽¹⁾	TL081C, TL082C, TL084C			TL081AC, TL082AC, TL084AC			TL081BC, TL082BC, TL084BC			TL081I, TL082I, TL084I			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
V_{IO} Input offset voltage	$V_O = 0$, $R_S = 50\ \Omega$	25°C		3	15		3	6		2	3		3	6	mV
		Full range			20			7.5			5			9	
α_{VIO} Temperature coefficient of input offset voltage	$V_O = 0$, $R_S = 50\ \Omega$	Full range		18			18			18			18		$\mu\text{V}/^\circ\text{C}$
I_{IO} Input offset current ⁽²⁾	$V_O = 0$	25°C		5	200		5	100		5	100		5	100	pA
		Full range			2			2			2			10	nA
I_{IB} Input bias current ⁽²⁾	$V_O = 0$	25°C		30	400		30	200		30	200		30	200	pA
		Full range			10			7			7			20	nA
V_{ICR} Common-mode input voltage range		25°C	± 11	–12 to 15		± 11	–12 to 15		± 11	–12 to 15		± 11	–12 to 15		V
V_{OM} Maximum peak output voltage swing	$R_L = 10\text{ k}\Omega$	25°C	± 12	± 13.5		± 12	± 13.5		± 12	± 13.5		± 12	± 13.5		V
	$R_L \geq 10\text{ k}\Omega$		± 12			± 12			± 12			± 12			
	$R_L \geq 2\text{ k}\Omega$	Full range	± 10	± 12		± 10	± 12		± 10	± 12		± 10	± 12		
A_{VD} Large-signal differential voltage amplification	$V_O = \pm 10\text{ V}$, $R_L \geq 2\text{ k}\Omega$	25°C	25	200		50	200		50	200		50	200		V/mV
		Full range	15			15			25			25			
B_1 Unity-gain bandwidth		25°C		3			3			3			3		MHz
r_i Input resistance		25°C		10^{12}			10^{12}			10^{12}			10^{12}		Ω
CMRR Common-mode rejection ratio	$V_{IC} = V_{ICRmin}$, $V_O = 0$, $R_S = 50\ \Omega$	25°C	70	86		75	86		75	86		75	86		dB
k_{SVR} Supply-voltage rejection ratio ($\Delta V_{CC\pm}/\Delta V_{IO}$)	$V_{CC} = \pm 15\text{ V}$ to $\pm 9\text{ V}$, $V_O = 0$, $R_S = 50\ \Omega$	25°C	70	86		80	86		80	86		80	86		dB
I_{CC} Supply current (each amplifier)	$V_O = 0$, No load	25°C		1.4	2.8		1.4	2.8		1.4	2.8		1.4	2.8	mA
V_{O1}/V_{O2} Crosstalk attenuation	$A_{VD} = 100$	25°C		120			120			120			120		dB

- (1) All characteristics are measured under open-loop conditions with zero common-mode voltage, unless otherwise specified. Full range for T_A is 0°C to 70°C for TL08_C, TL08_AC, TL08_BC and –40°C to 85°C for TL08_I.
- (2) Input bias currents of an FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive, as shown in Figure 6-52. Pulse techniques must be used that maintain the junction temperature as close to the ambient temperature as possible.

6.13 Electrical Characteristics for TL08xM and TL084x

$V_{CC\pm} = \pm 15\text{ V}$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS ⁽¹⁾	T_A	TL081M, TL082M			TL084Q, TL084M			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
V_{IO} Input offset voltage	$V_O = 0$, $R_S = 50\ \Omega$	25°C		3	6		3	9	mV
		Full range			9			15	
α_{VIO} Temperature coefficient of input offset voltage	$V_O = 0$, $R_S = 50\ \Omega$	Full range		18			18		$\mu\text{V}/^\circ\text{C}$
I_{IO} Input offset current ⁽²⁾	$V_O = 0$	25°C		5	100		5	100	pA
		125°C			20			20	nA
I_{IB} Input bias current ⁽²⁾	$V_O = 0$	25°C		30	200		30	200	pA
		125°C			50			50	nA
V_{ICR} Common-mode input voltage range		25°C	± 11	-12 to 15		± 11	-12 to 15		V
V_{OM} Maximum peak output voltage swing	$R_L = 10\text{ k}\Omega$	25°C	± 12	± 13.5		± 12	± 13.5		V
	$R_L \geq 10\text{ k}\Omega$	Full range	± 12			± 12			
	$R_L \geq 2\text{ k}\Omega$		± 10	± 12		± 10	± 12		
A_{VD} Large-signal differential voltage amplification	$V_O = \pm 10\text{ V}$, $R_L \geq 2\text{ k}\Omega$	25°C	25	200		25	200		V/mV
		Full range	15			15			
B_1 Unity-gain bandwidth		25°C		3			3		MHz
r_i Input resistance		25°C		10^{12}			10^{12}		Ω
CMRR Common-mode rejection ratio	$V_{IC} = V_{ICRmin}$, $V_O = 0$, $R_S = 50\ \Omega$	25°C	80	86		80	86		dB
k_{SVR} Supply-voltage rejection ratio ($\Delta V_{CC\pm}/\Delta V_{IO}$)	$V_{CC} = \pm 15\text{ V}$ to $\pm 9\text{ V}$, $V_O = 0$, $R_S = 50\ \Omega$	25°C	80	86		80	86		dB
I_{CC} Supply current (each amplifier)	$V_O = 0$, No load	25°C		1.4	2.8		1.4	2.8	mA
V_{O1}/V_{O2} Crosstalk attenuation	$A_{VD} = 100$	25°C		120			120		dB

(1) All characteristics are measured under open-loop conditions, with zero common-mode input voltage, unless otherwise specified.

(2) Input bias currents of a FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive, as shown in Figure 6-52. Pulse techniques must be used that maintain the junction temperatures as close to the ambient temperature as possible.

6.14 Switching Characteristics

 $V_{CC\pm} = \pm 15\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
SR Slew rate at unity gain	$V_I = 10\text{ V}$, $R_L = 2\text{ k}\Omega$, $C_L = 100\text{ pF}$, see Figure 7-1	8 ⁽¹⁾	13		V/ μs
	$V_I = 10\text{ V}$, $R_L = 2\text{ k}\Omega$, $C_L = 100\text{ pF}$, $T_A = -55^\circ\text{C}$ to 125°C , see Figure 7-1	5 ⁽¹⁾			
t_r Rise-time	$V_I = 20\text{ V}$, $R_L = 2\text{ k}\Omega$, $C_L = 100\text{ pF}$, see Figure 7-1		0.05		μs
overshoot factor			20%		
V_n Equivalent input noise voltage	$R_S = 20\text{ }\Omega$	$f = 1\text{ kHz}$		18	nV/ $\sqrt{\text{Hz}}$
		$f = 10\text{ Hz}$ to 10 kHz		4	μV
I_n Equivalent input noise current	$R_S = 20\text{ }\Omega$	$f = 1\text{ kHz}$		0.01	pA/ $\sqrt{\text{Hz}}$
THD Total harmonic distortion	$V_{I\text{rms}} = 6\text{ V}$, $A_{VD} = 1$, $R_S \leq 1\text{ k}\Omega$, $R_L \geq 2\text{ k}\Omega$, $f = 1\text{ kHz}$		0.003%		

(1) On products compliant to MIL-PRF-38535, this parameter is not production tested.

6.15 Dissipation Rating Table

PACKAGE	$T_A \leq 25^\circ\text{C}$ POWER RATING	DERATING FACTOR	DERATE ABOVE T_A	$T_A = 70^\circ\text{C}$ POWER RATING	$T_A = 85^\circ\text{C}$ POWER RATING	$T_A = 125^\circ\text{C}$ POWER RATING
D (14 pin)	680 mW	7.6 mW/ $^\circ\text{C}$	60°C	604 mW	490 mW	186 mW
FK	680 mW	11.0 mW/ $^\circ\text{C}$	88°C	680 mW	680 mW	273 mW
J	680 mW	11.0 mW/ $^\circ\text{C}$	88°C	680 mW	680 mW	273 mW
JG	680 mW	8.4 mW/ $^\circ\text{C}$	69°C	672 mW	546 mW	210 mW

6.16 Typical Characteristics: TL08xH

at $T_A = 25^\circ\text{C}$, $V_S = 40\text{ V}$ ($\pm 20\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$, and $C_L = 20\text{ pF}$ (unless otherwise noted)

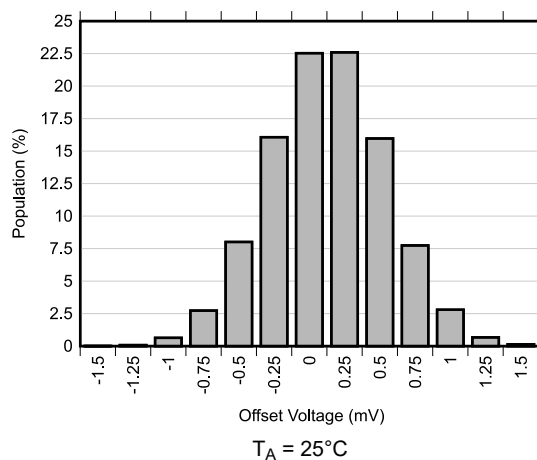


Figure 6-1. Offset Voltage Production Distribution

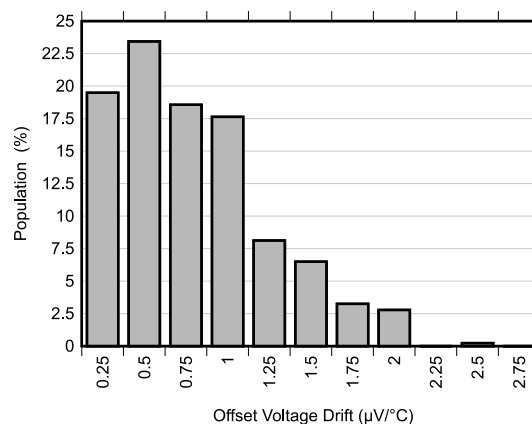


Figure 6-2. Offset Voltage Drift Distribution

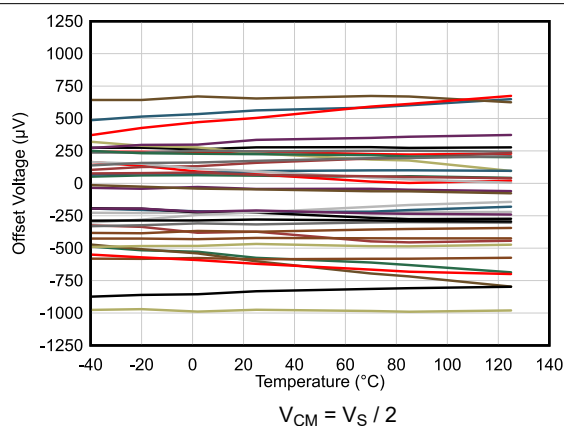


Figure 6-3. Offset Voltage vs Temperature

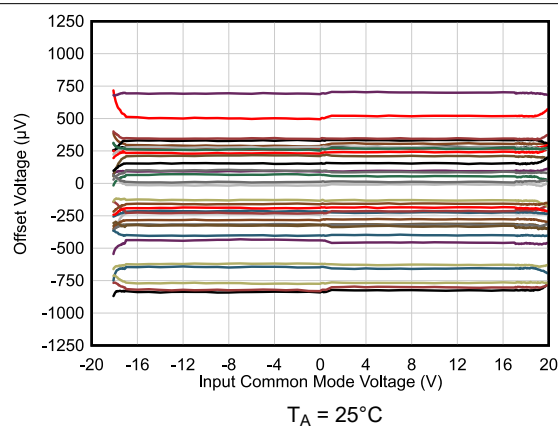


Figure 6-4. Offset Voltage vs Common-Mode Voltage

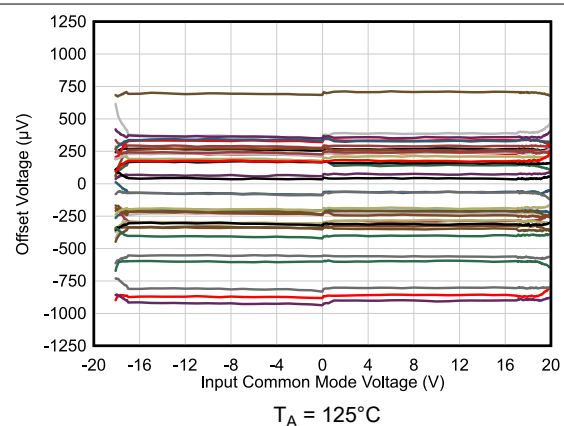


Figure 6-5. Offset Voltage vs Common-Mode Voltage

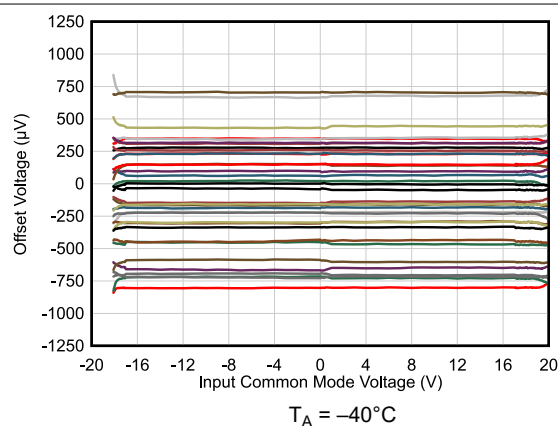


Figure 6-6. Offset Voltage vs Common-Mode Voltage

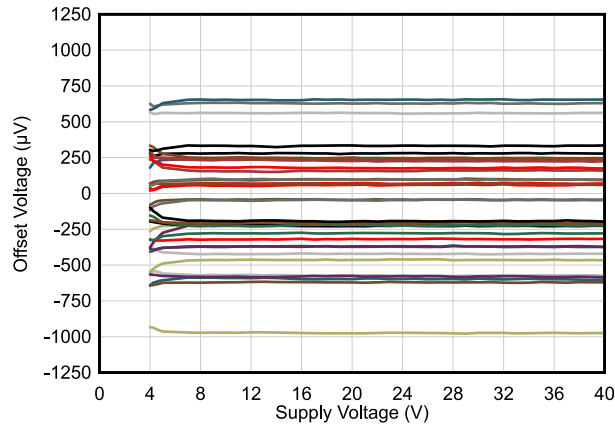


Figure 6-7. Offset Voltage vs Power Supply

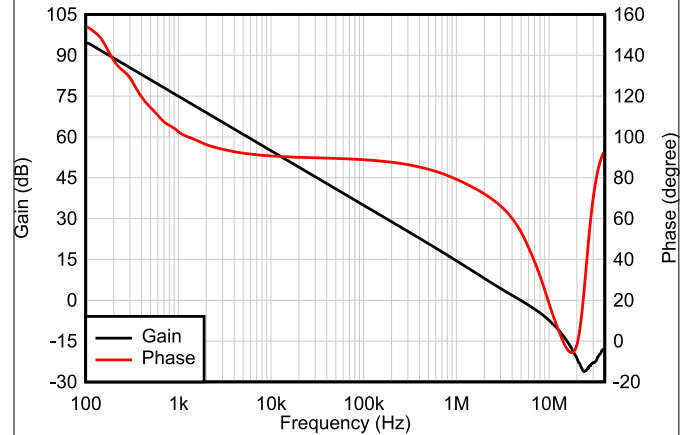


Figure 6-8. Open-Loop Gain and Phase vs Frequency

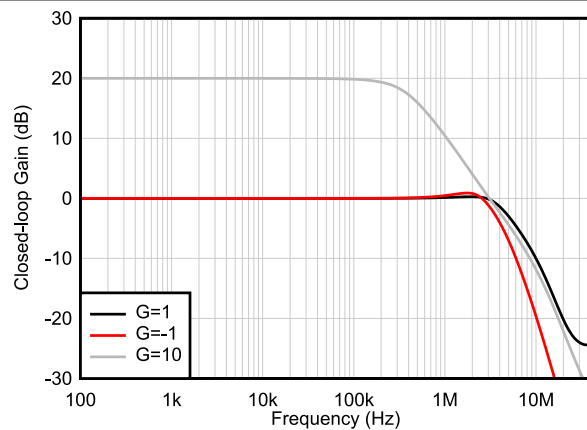


Figure 6-9. Closed-Loop Gain vs Frequency

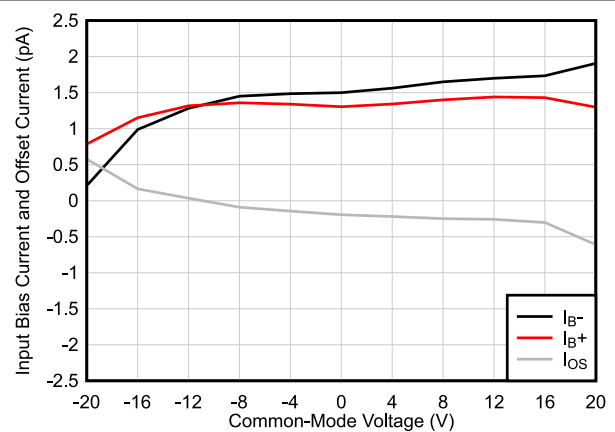


Figure 6-10. Input Bias Current vs Common-Mode Voltage

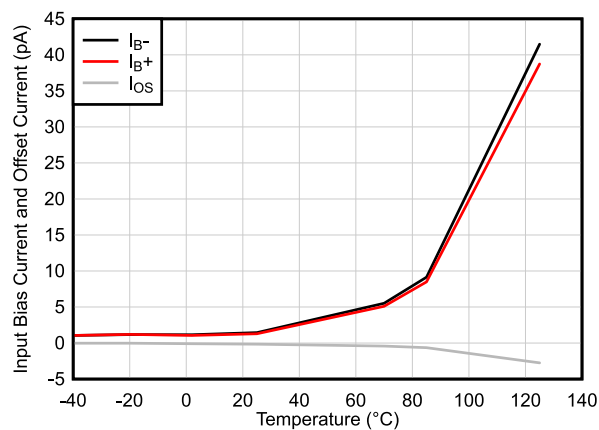


Figure 6-11. Input Bias Current vs Temperature

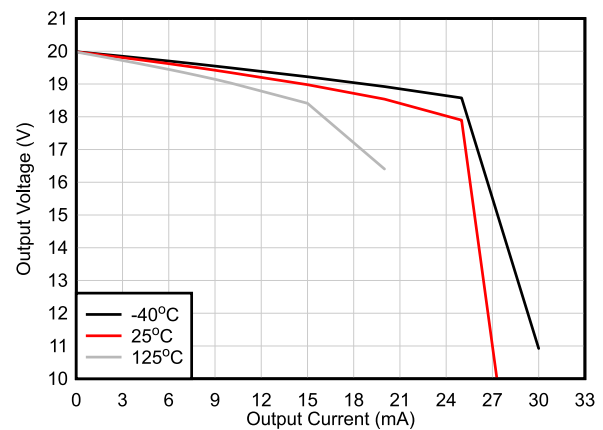


Figure 6-12. Output Voltage Swing vs Output Current (Sourcing)

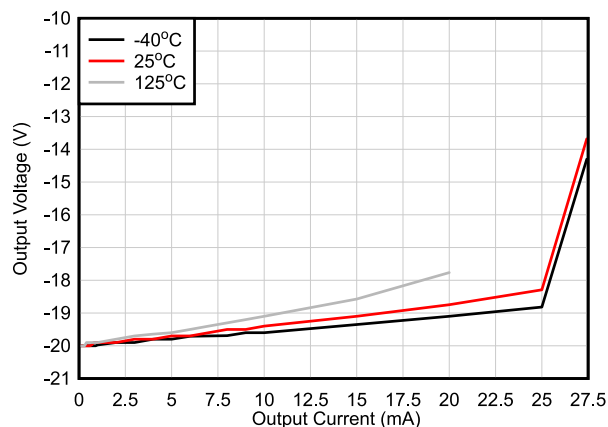


Figure 6-13. Output Voltage Swing vs Output Current (Sinking)

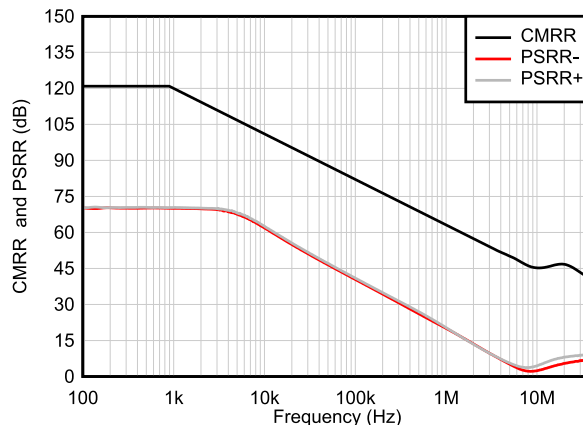


Figure 6-14. CMRR and PSRR vs Frequency

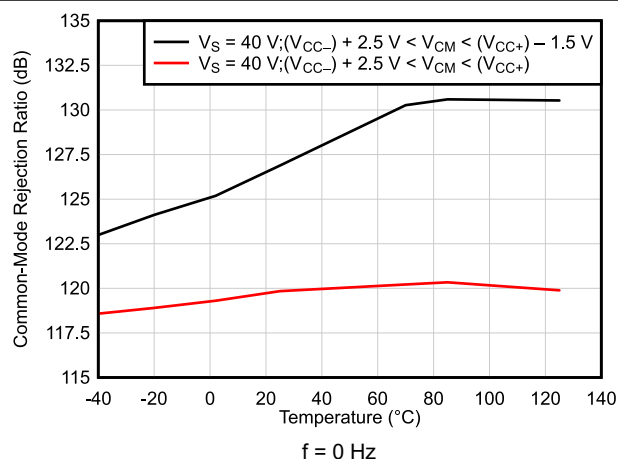


Figure 6-15. CMRR vs Temperature (dB)

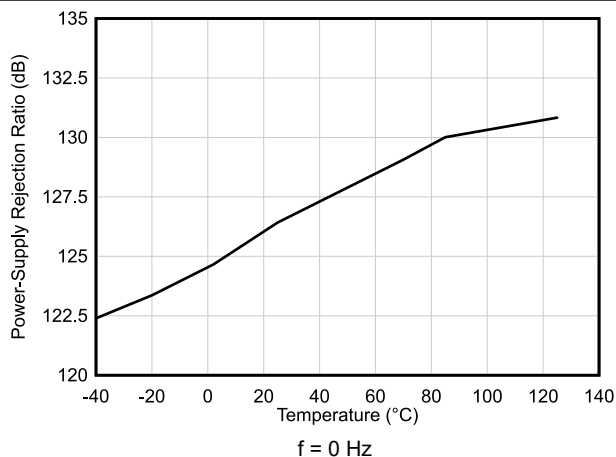


Figure 6-16. PSRR vs Temperature (dB)

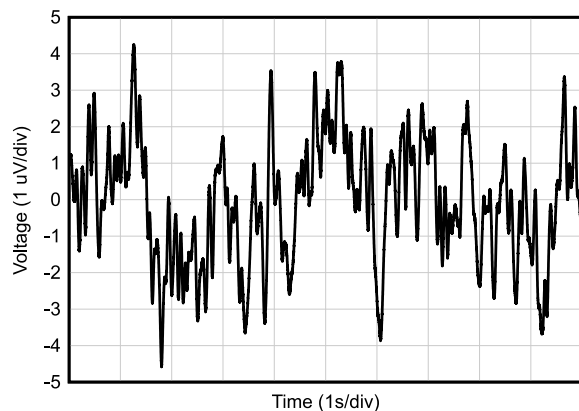


Figure 6-17. 0.1-Hz to 10-Hz Noise

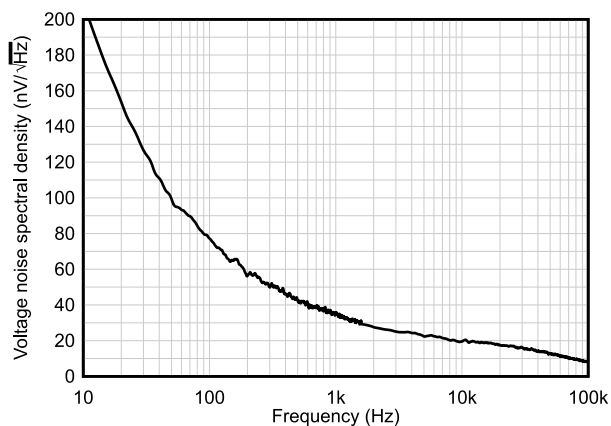


Figure 6-18. Input Voltage Noise Spectral Density vs Frequency

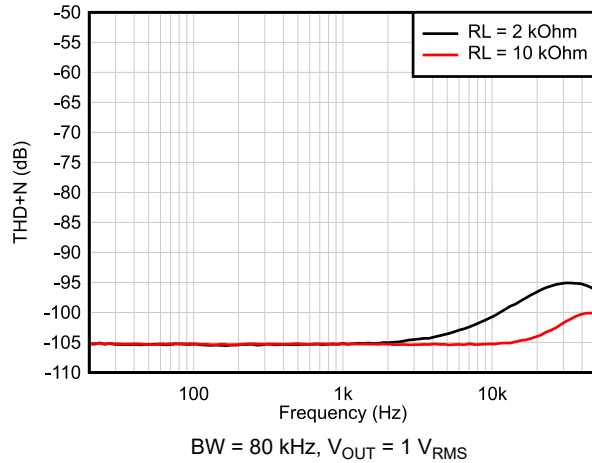


Figure 6-19. THD+N Ratio vs Frequency

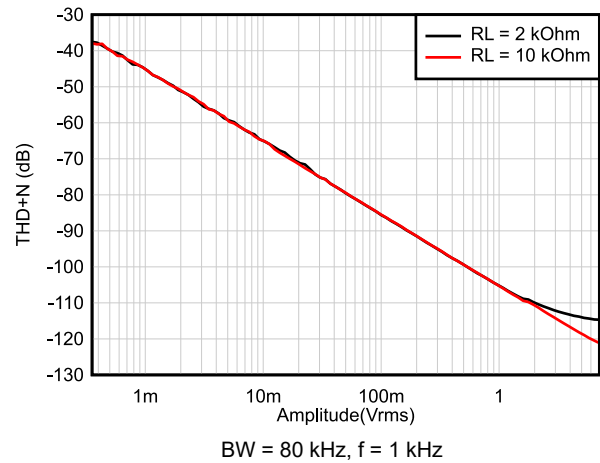


Figure 6-20. THD+N vs Output Amplitude

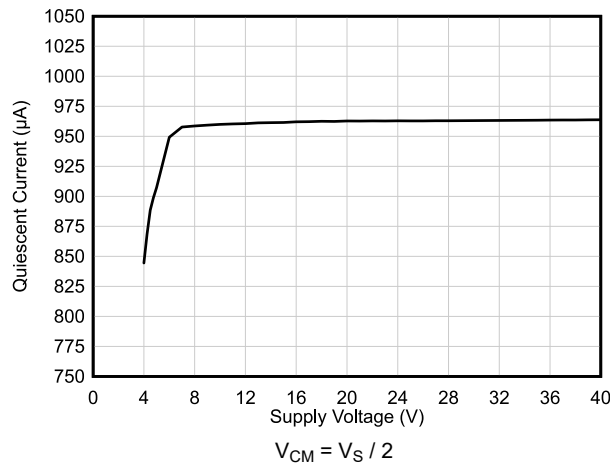


Figure 6-21. Quiescent Current vs Supply Voltage

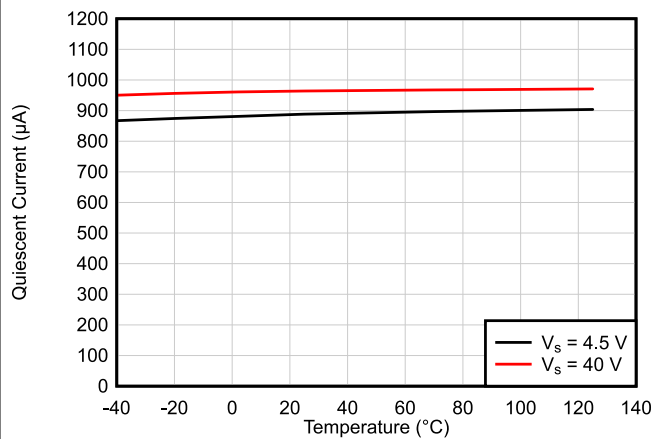


Figure 6-22. Quiescent Current vs Temperature

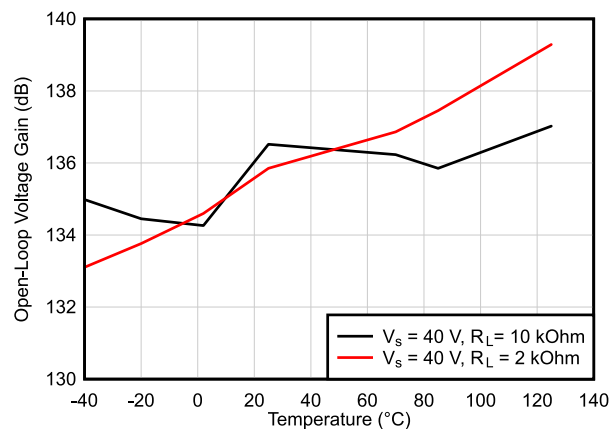


Figure 6-23. Open-Loop Voltage Gain vs Temperature (dB)

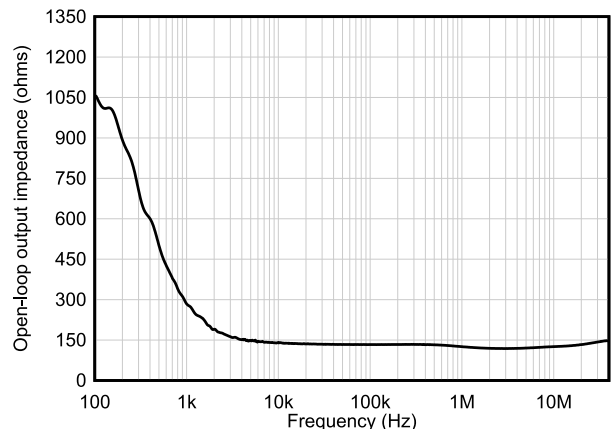


Figure 6-24. Open-Loop Output Impedance vs Frequency

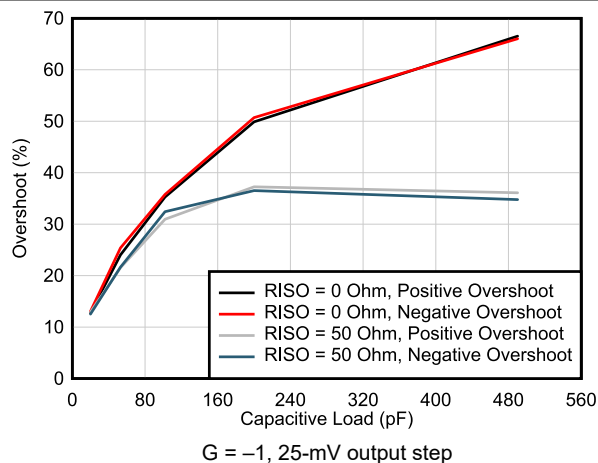


Figure 6-25. Small-Signal Overshoot vs Capacitive Load

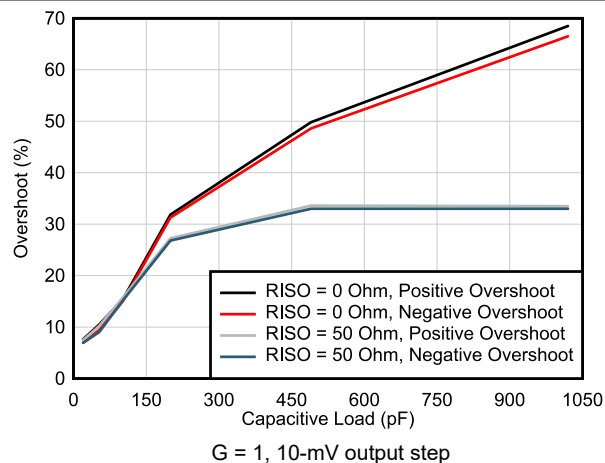


Figure 6-26. Small-Signal Overshoot vs Capacitive Load

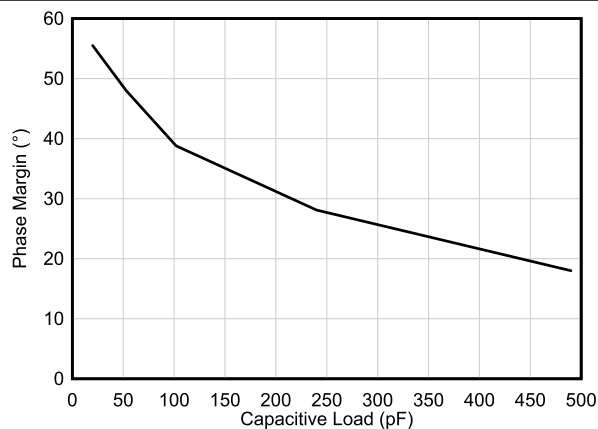


Figure 6-27. Phase Margin vs Capacitive Load

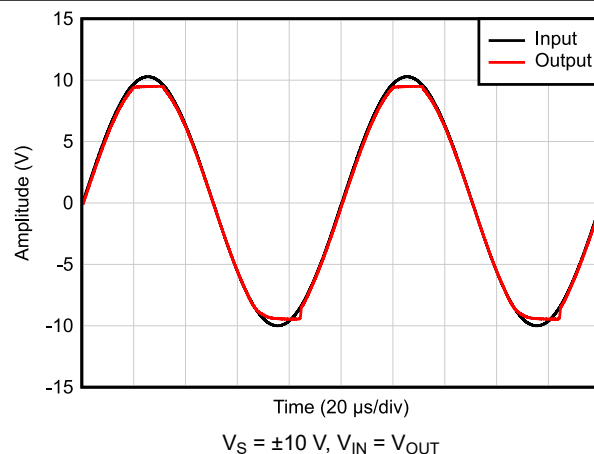


Figure 6-28. No Phase Reversal

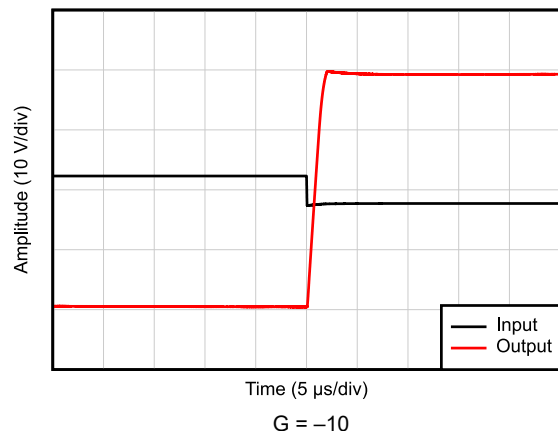


Figure 6-29. Positive Overload Recovery

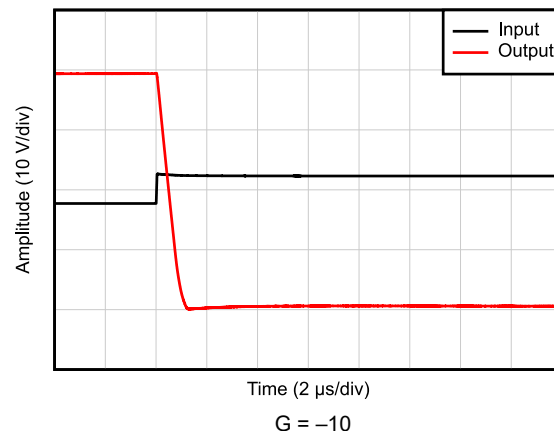
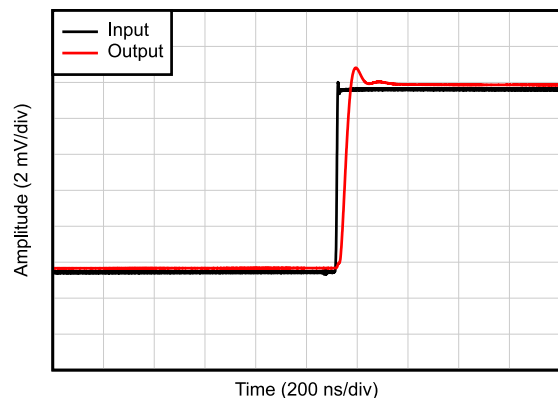
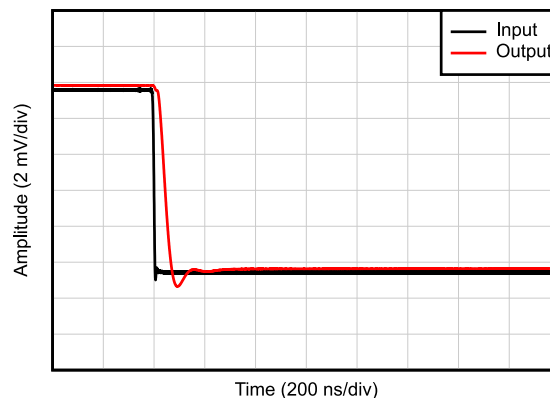


Figure 6-30. Negative Overload Recovery



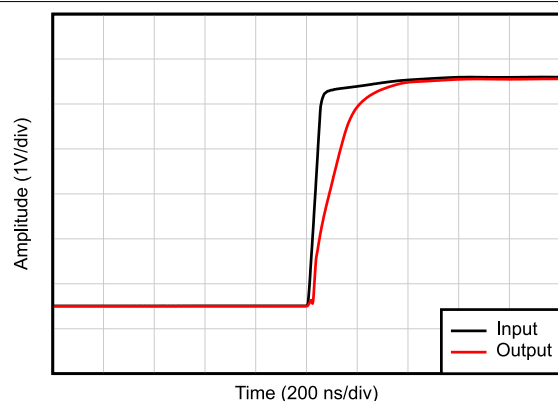
$C_L = 20$ pF, $G = 1$, 10-mV step response

Figure 6-31. Small-Signal Step Response, Rising



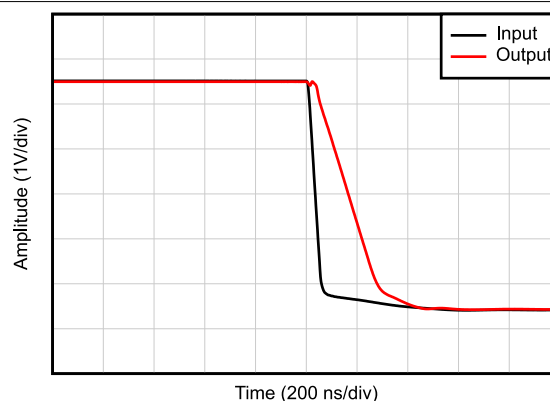
$C_L = 20$ pF, $G = 1$, 10-mV step response

Figure 6-32. Small-Signal Step Response, Falling



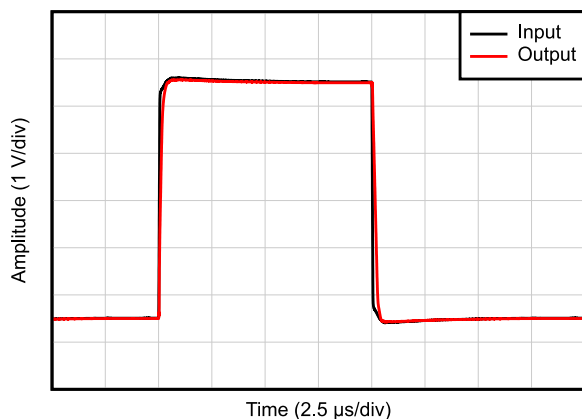
$C_L = 20$ pF, $G = 1$

Figure 6-33. Large-Signal Step Response (Rising)



$C_L = 20$ pF, $G = 1$

Figure 6-34. Large-Signal Step Response (Falling)



$C_L = 20$ pF, $G = 1$

Figure 6-35. Large-Signal Step Response

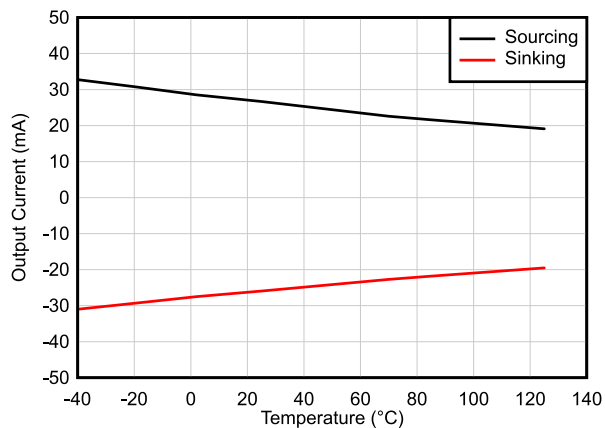


Figure 6-36. Short-Circuit Current vs Temperature

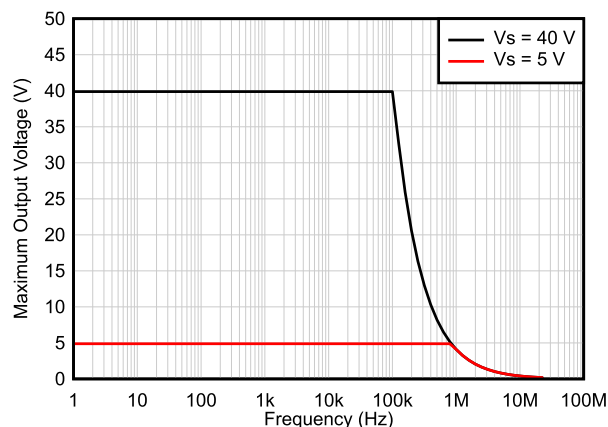


Figure 6-37. Maximum Output Voltage vs Frequency

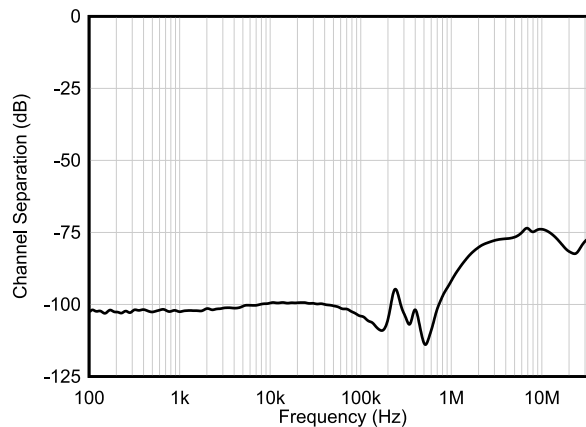


Figure 6-38. Channel Separation vs Frequency

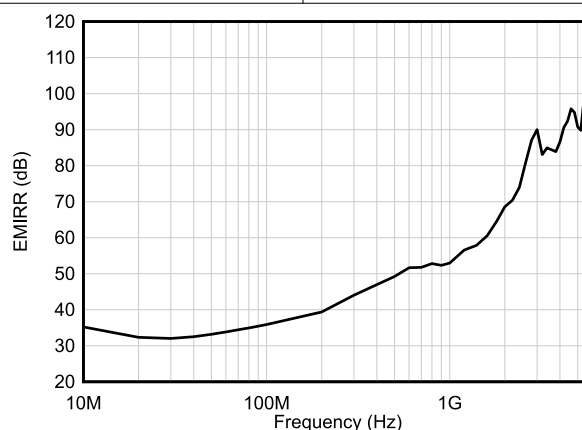


Figure 6-39. EMIRR (Electromagnetic Interference Rejection Ratio) vs Frequency

6.17 Typical Characteristics: All Other Devices

Data at high and low temperatures are applicable only within the rated operating free-air temperature ranges of the various devices. The Figure numbers referenced in the following graphs are located in [Section 7](#).

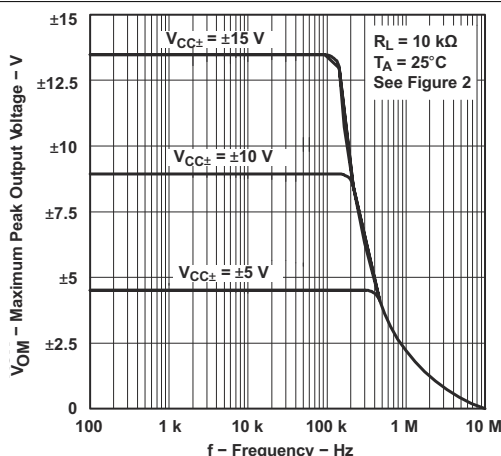


Figure 6-40. Maximum Peak Output Voltage vs Frequency

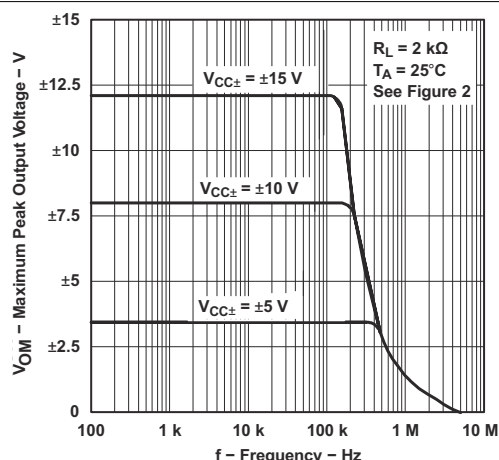


Figure 6-41. Maximum Peak Output Voltage vs Frequency

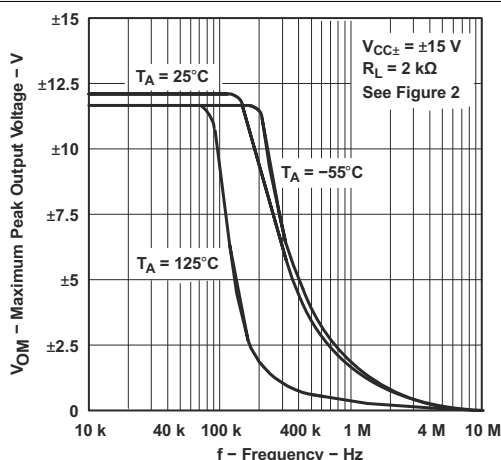


Figure 6-42. Maximum Peak Output Voltage vs Frequency

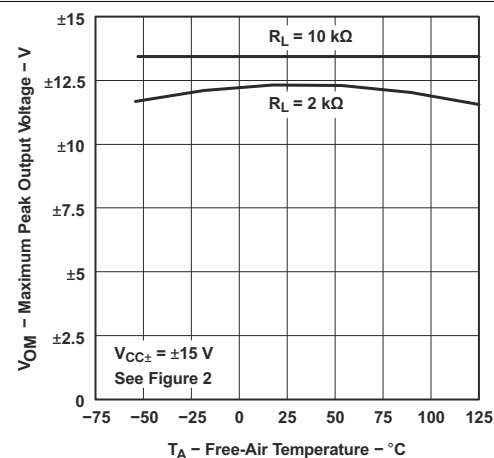


Figure 6-43. Maximum Peak Output Voltage vs Free-Air Temperature

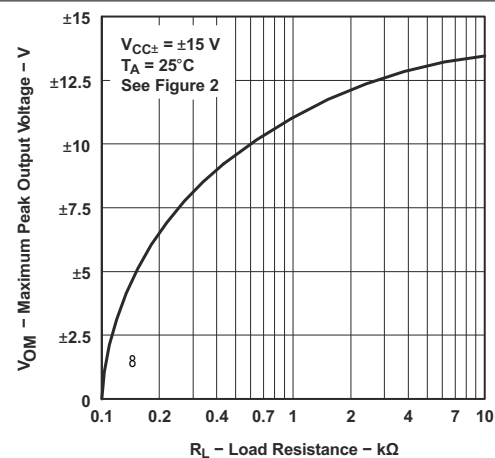


Figure 6-44. Maximum Peak Output Voltage vs Load Resistance

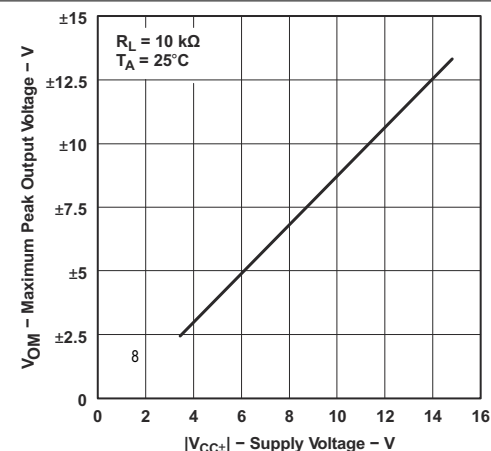


Figure 6-45. Maximum Peak Output Voltage vs Supply Voltage

6.17 Typical Characteristics: All Other Devices (continued)

Data at high and low temperatures are applicable only within the rated operating free-air temperature ranges of the various devices. The Figure numbers referenced in the following graphs are located in [Section 7](#).

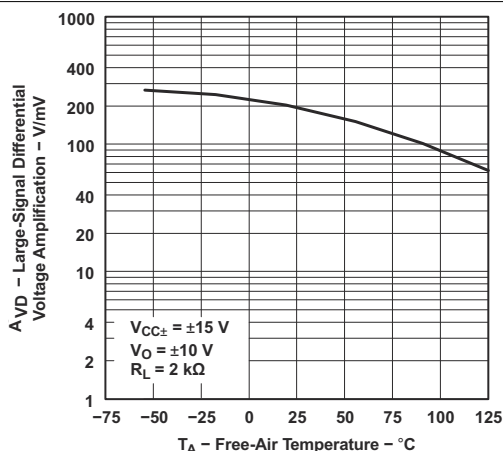


Figure 6-46. Large-Signal Differential Voltage Amplification vs Free-Air Temperature

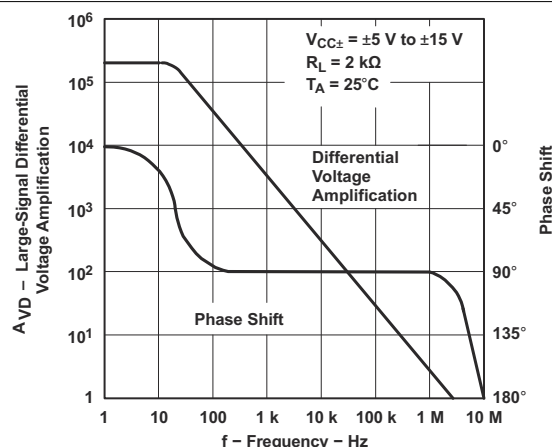


Figure 6-47. Large-Signal Differential Voltage Amplification and Phase Shift vs Frequency

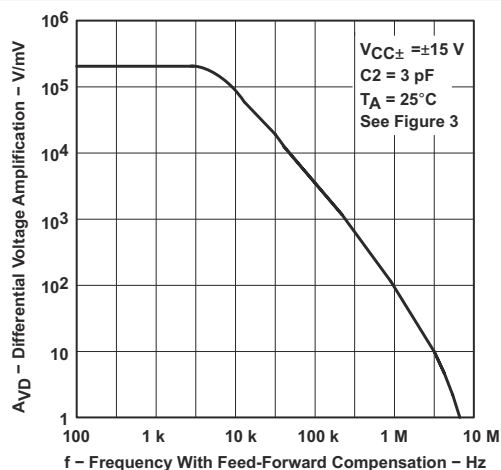


Figure 6-48. Differential Voltage Amplification vs Frequency with Feed-Forward Compensation

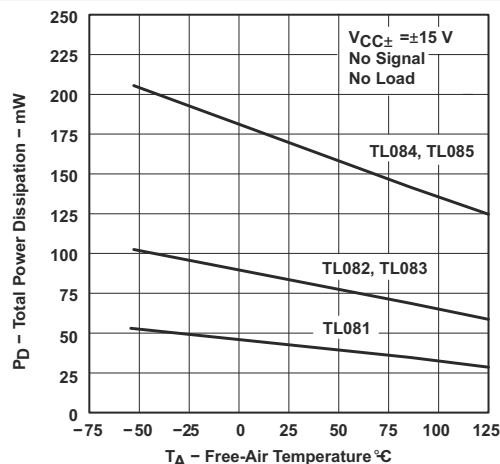


Figure 6-49. Total Power Dissipation vs Free-Air Temperature

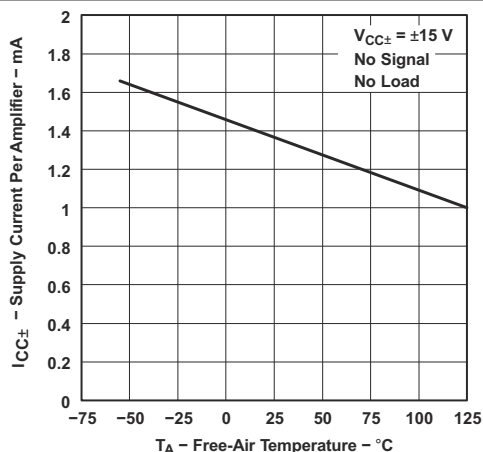


Figure 6-50. Supply Current per Amplifier vs Free-Air Temperature

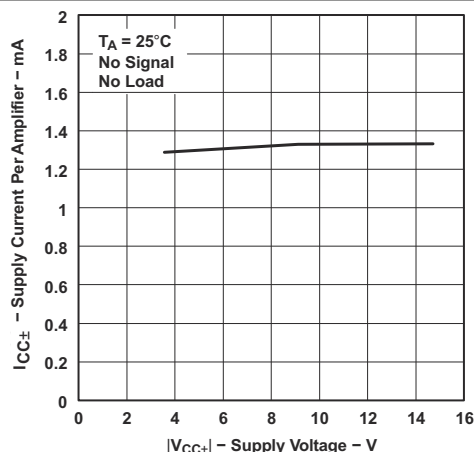


Figure 6-51. Supply Current per Amplifier vs Supply Voltage

6.17 Typical Characteristics: All Other Devices (continued)

Data at high and low temperatures are applicable only within the rated operating free-air temperature ranges of the various devices. The Figure numbers referenced in the following graphs are located in [Section 7](#).

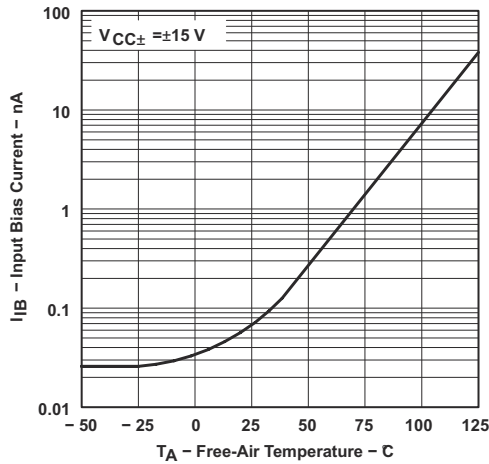


Figure 6-52. Input Bias Current vs Free-Air Temperature

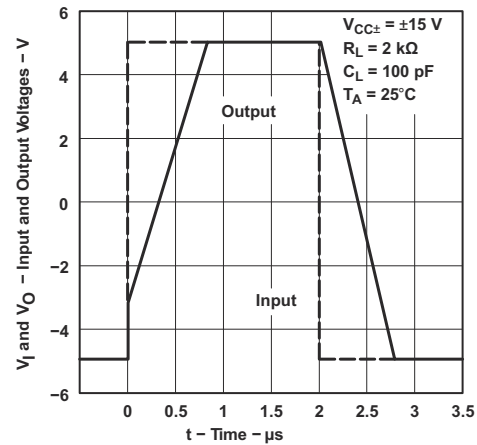


Figure 6-53. Voltage-Follower Large-Signal Pulse Response

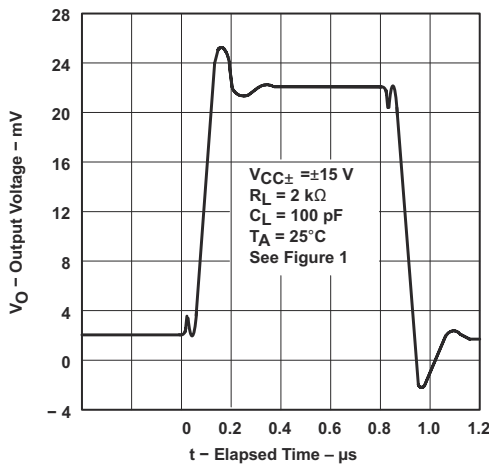


Figure 6-54. Output Voltage vs Elapsed Time

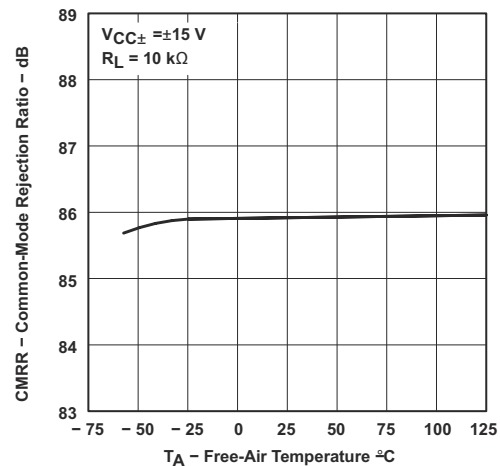


Figure 6-55. Common-Mode Rejection Ratio vs Free-Air Temperature

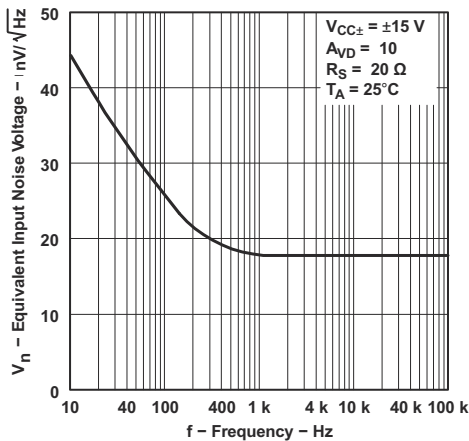


Figure 6-56. Equivalent Input Noise Voltage vs Frequency

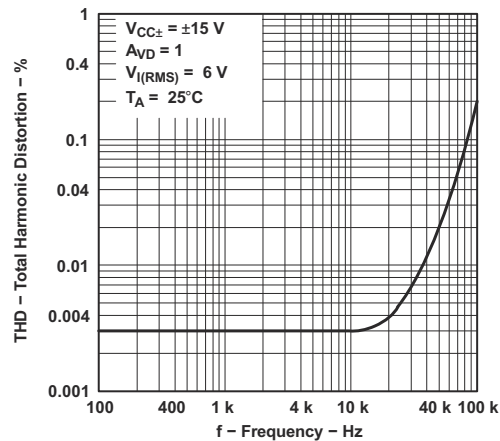


Figure 6-57. Total Harmonic Distortion vs Frequency

7 Parameter Measurement Information

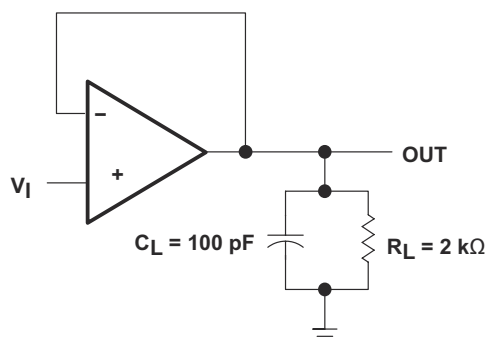


Figure 7-1. Test Figure 1

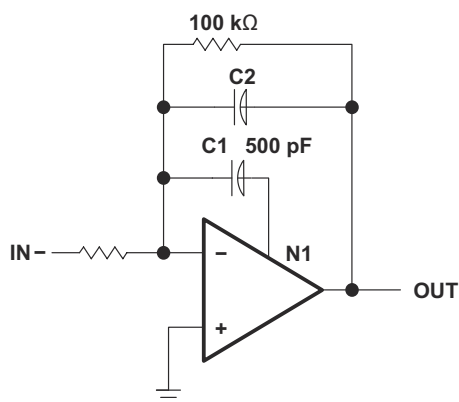


Figure 7-3. Test Figure 3

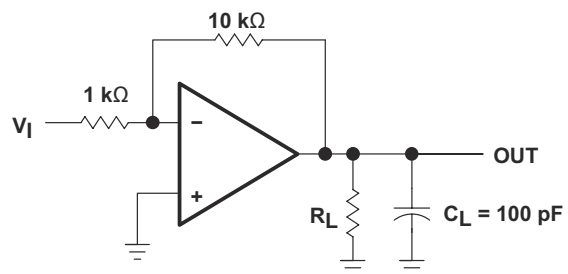


Figure 7-2. Test Figure 2

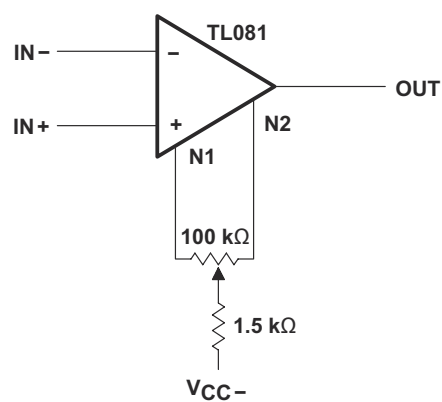


Figure 7-4. Test Figure 4

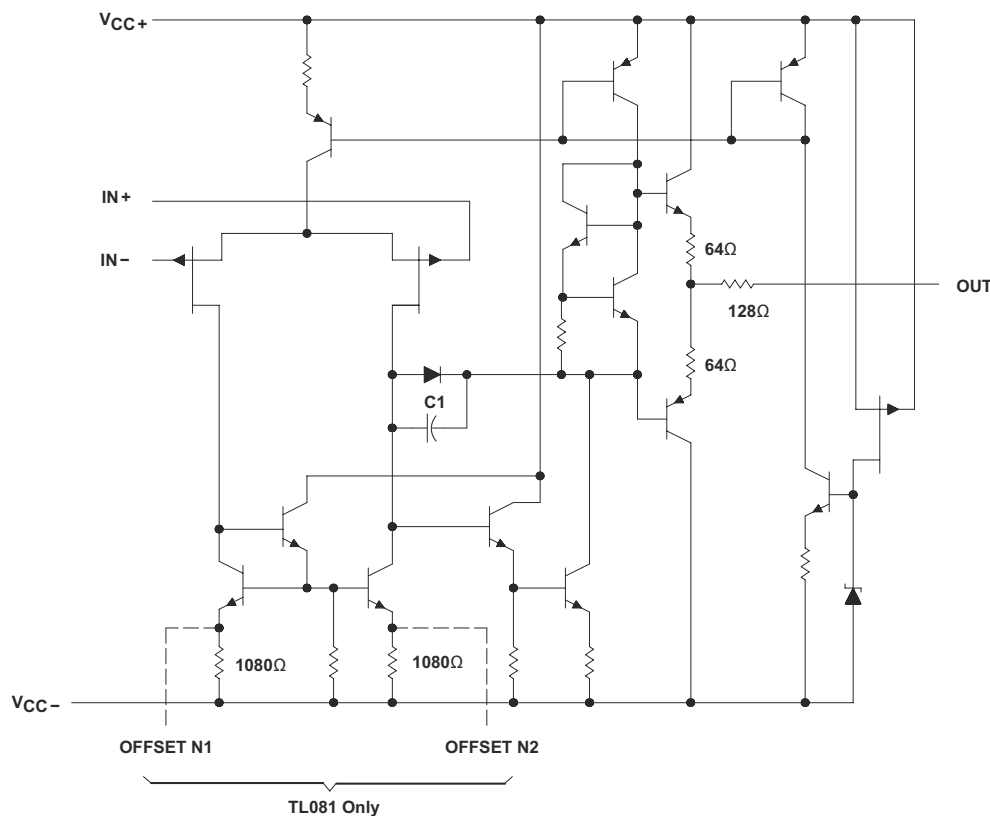
8 Detailed Description

8.1 Overview

The TL08xH family (TL081H, TL082H, and TL084H) is the next-generation family of the industry standard TL08x (TL081, TL082, and TL084) high-voltage general purpose amplifiers. These devices provide outstanding value for cost-sensitive applications requiring high slew rate with high voltage signals, such as motor drive and inverter systems.

A robust MUX-friendly input stage enhances flexibility in design, with common-mode voltage range extending to the positive rail as well as improved settling time in multi-channel applications. Low offset voltage (1 mV, typ) and low offset voltage drift ($2 \mu\text{V}/^\circ\text{C}$) allows the TL08xH family to be used in rugged applications requiring precision current and voltage sensing. High voltage operation (up to 40 V) and high slew rate ($20 \text{ V}/\mu\text{s}$) make the TL08xH family a premier choice for high-voltage applications with fast transients.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Total Harmonic Distortion

Harmonic distortions to an audio signal are created by electronic components in a circuit. Total harmonic distortion (THD) is a measure of harmonic distortions accumulated by a signal in an audio system. These devices have a very low THD of 0.003% meaning that the TL08x devices will add little harmonic distortion when used in audio signal applications.

8.3.2 Slew Rate

The slew rate is the rate at which an operational amplifier can change its output when there is a change on the input. These devices have a $13\text{-V}/\mu\text{s}$ slew rate.

8.4 Device Functional Modes

These devices are powered on when the supply is connected. This device can be operated as a single-supply operational amplifier or dual-supply amplifier depending on the application.

9 Applications and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

9.1 Application Information

The TL08x series of operational amplifiers can be used in countless applications. The few applications in this section show principles used in all applications of these parts.

9.2 Typical Applications

9.2.1 Inverting Amplifier Application

A typical application for an operational amplifier is an inverting amplifier. This amplifier takes a positive voltage on the input, and makes it a negative voltage of the same magnitude. In the same manner, it also makes negative voltages positive.

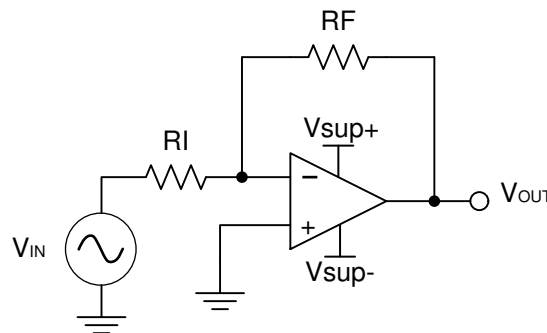


Figure 9-1. Schematic for Inverting Amplifier Application

9.2.1.1 Design Requirements

The supply voltage must be chosen such that it is larger than the input voltage range and output range. For instance, this application will scale a signal of ± 0.5 V to ± 1.8 V. Setting the supply at ± 12 V is sufficient to accommodate this application.

9.2.1.2 Detailed Design Procedure

Determine the gain required by the inverting amplifier:

$$A_V = \frac{V_{OUT}}{V_{IN}} \quad (1)$$

$$A_V = \frac{1.8}{-0.5} = -3.6 \quad (2)$$

Once the desired gain is determined, choose a value for R_I or R_F . Choosing a value in the $k\Omega$ range is desirable because the amplifier circuit will use currents in the milliamp range. This ensures the part will not draw too much current. This example will choose $10\text{ k}\Omega$ for R_I which means $36\text{ k}\Omega$ will be used for R_F . This was determined by Equation 3.

$$A_V = -\frac{R_F}{R_I} \quad (3)$$

9.2.1.3 Application Curve

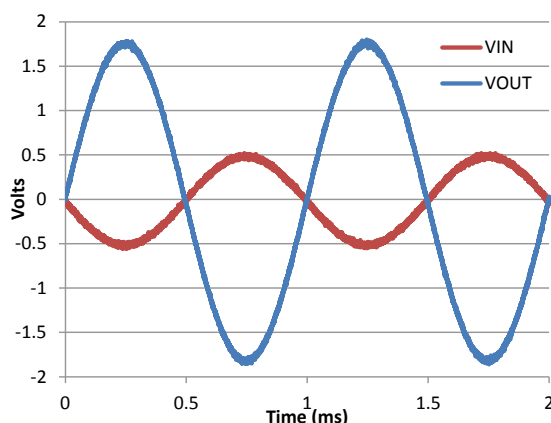


Figure 9-2. Input and Output Voltages of the Inverting Amplifier

9.3 System Examples

9.3.1 General Applications

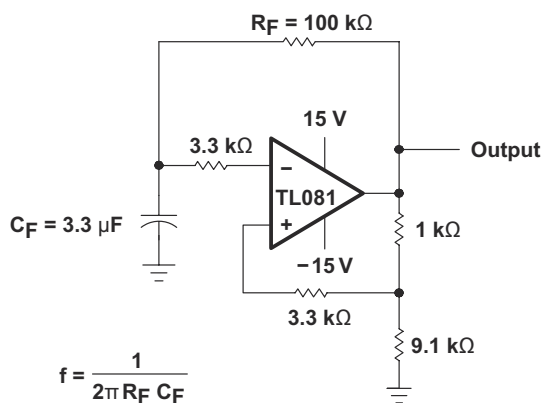


Figure 9-3. 0.5-Hz Square-Wave Oscillator

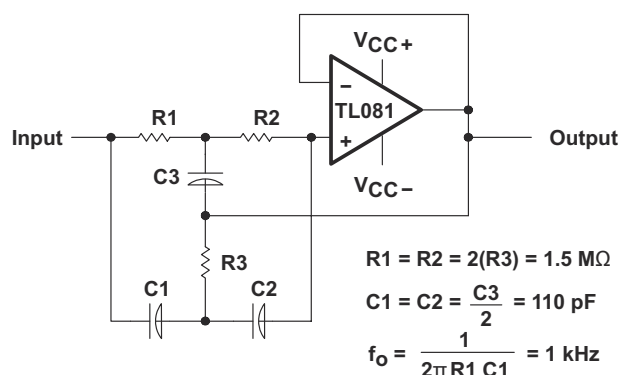


Figure 9-4. High-Q Notch Filter

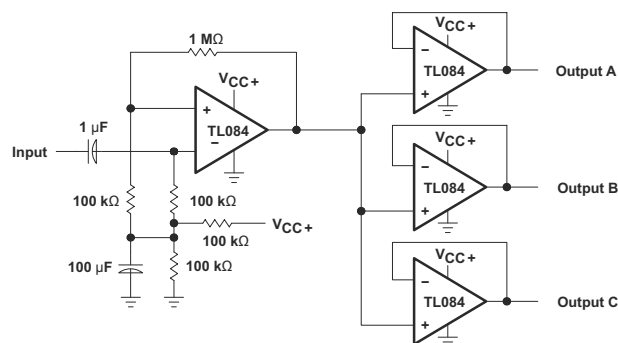
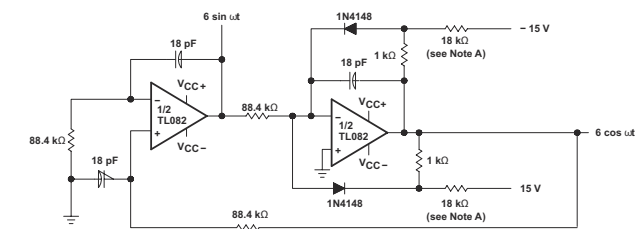


Figure 9-5. Audio-Distribution Amplifier



A. These resistor values may be adjusted for a symmetrical output.

Figure 9-6. 100-kHz Quadrature Oscillator

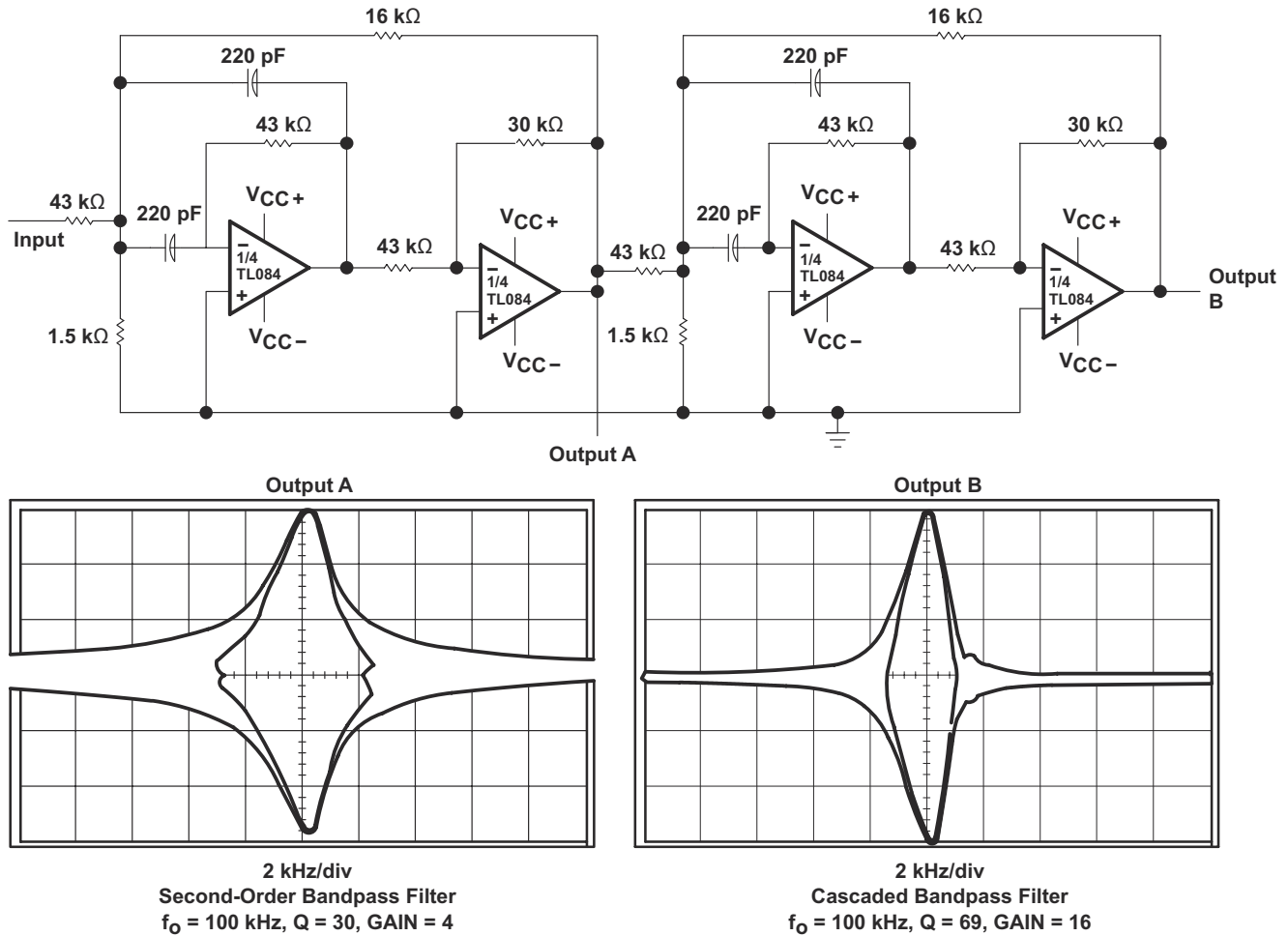


Figure 9-7. Positive-Feedback Bandpass Filter

10 Power Supply Recommendations

CAUTION

Supply voltages larger than 36 V for a single-supply or outside the range of ± 18 V for a dual-supply can permanently damage the device (see [Section 6.2](#)).

Place 0.1- μ F bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high impedance power supplies. For more detailed information on bypass capacitor placement, refer to [Section 11](#).

11 Layout

11.1 Layout Guidelines

For best operational performance of the device, use good PCB layout practices, including:

- Noise can propagate into analog circuitry through the power pins of the circuit as a whole, as well as the operational amplifier. Bypass capacitors are used to reduce the coupled noise by providing low impedance power sources local to the analog circuitry.
 - Connect low-ESR, 0.1- μ F ceramic bypass capacitors between each supply pin and ground, placed as close to the device as possible. A single bypass capacitor from V+ to ground is applicable for single-supply applications.
- Separate grounding for analog and digital portions of circuitry is one of the simplest and most-effective methods of noise suppression. One or more layers on multilayer PCBs are usually devoted to ground planes. A ground plane helps distribute heat and reduces EMI noise pickup. Make sure to physically separate digital and analog grounds, paying attention to the flow of the ground current.
- To reduce parasitic coupling, run the input traces as far away from the supply or output traces as possible. If it is not possible to keep them separate, it is much better to cross the sensitive trace perpendicular as opposed to in parallel with the noisy trace.
- Place the external components as close to the device as possible. Keeping RF and RG close to the inverting input minimizes parasitic capacitance, as shown in [Section 11.2](#).
- Keep the length of input traces as short as possible. Always remember that the input traces are the most sensitive part of the circuit.
- Consider a driven, low-impedance guard ring around the critical traces. A guard ring can significantly reduce leakage currents from nearby traces that are at different potentials.

11.2 Layout Examples

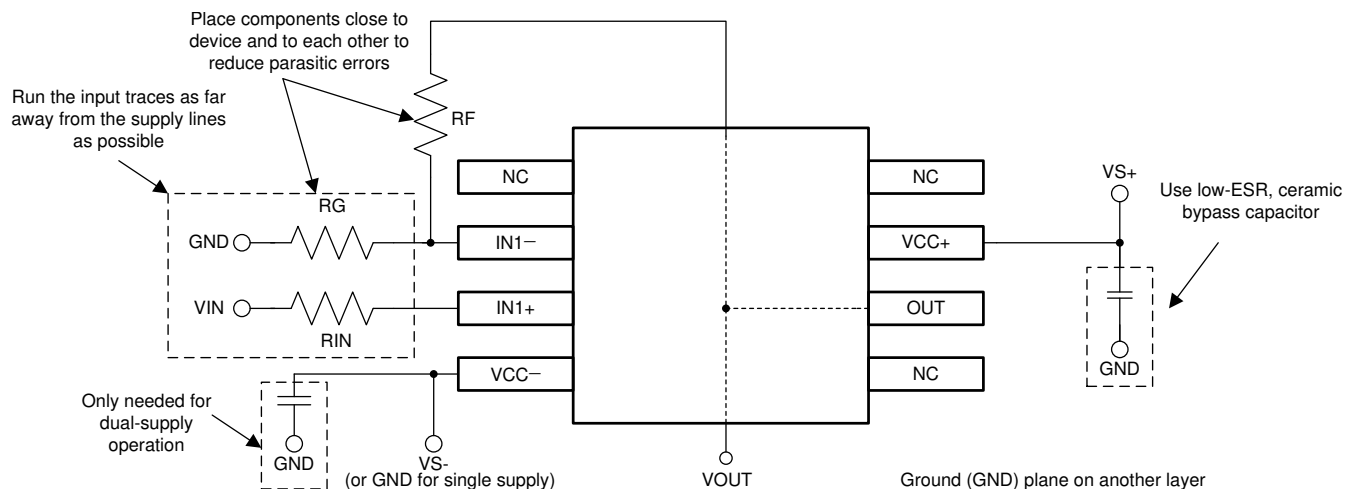


Figure 11-1. Operational Amplifier Board Layout for Noninverting Configuration

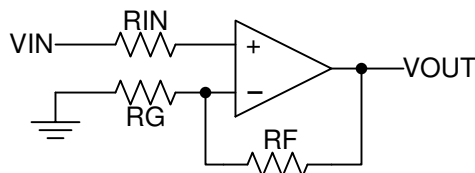


Figure 11-2. Operational Amplifier Schematic for Noninverting Configuration

12 Device and Documentation Support

12.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Subscribe to updates* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

12.2 Support Resources

TI E2E™ [support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

12.3 Trademarks

TI E2E™ is a trademark of Texas Instruments.

All trademarks are the property of their respective owners.

12.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

12.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
5962-9851501Q2A	ACTIVE	LCCC	FK	20	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962- 9851501Q2A TL082MFKB	Samples
5962-9851501QPA	ACTIVE	CDIP	JG	8	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	9851501QPA TL082M	Samples
5962-9851503Q2A	ACTIVE	LCCC	FK	20	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962- 9851503Q2A TL084 MFKB	Samples
5962-9851503QCA	ACTIVE	CDIP	J	14	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962-9851503QC A TL084MJB	Samples
TL081ACD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	081AC	
TL081ACDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	081AC	Samples
TL081ACP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL081ACP	Samples
TL081BCD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	081BC	
TL081BCDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	081BC	Samples
TL081BCP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL081BCP	Samples
TL081CD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL081C	
TL081CDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL081C	Samples
TL081CP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL081CP	Samples
TL081CPE4	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL081CP	Samples
TL081CPSR	ACTIVE	SO	PS	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T081	Samples
TL081HIDBVR	ACTIVE	SOT-23	DBV	5	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	T81V	Samples
TL081HIDCKR	ACTIVE	SC70	DCK	5	3000	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 125	11P	Samples
TL081HIDR	ACTIVE	SOIC	D	8	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL081D	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TL081ID	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL081I	
TL081IDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL081I	Samples
TL081IP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	TL081IP	Samples
TL082ACD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	
TL082ACDE4	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	
TL082ACDG4	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	
TL082ACDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	Samples
TL082ACDRE4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	Samples
TL082ACDRG4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082AC	Samples
TL082ACP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL082ACP	Samples
TL082ACPSR	ACTIVE	SO	PS	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082A	Samples
TL082BCD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082BC	
TL082BCDE4	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082BC	
TL082BCDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	082BC	Samples
TL082BCP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL082BCP	Samples
TL082BCPE4	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL082BCP	Samples
TL082CD	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL082C	
TL082CDE4	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL082C	
TL082CDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL082C	Samples
TL082CDRE4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL082C	Samples
TL082CDRG4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL082C	Samples
TL082CP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL082CP	Samples
TL082CPSR	ACTIVE	SO	PS	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TL082CPSRG4	ACTIVE	SO	PS	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082	Samples
TL082CPW	LIFEBUY	TSSOP	PW	8	150	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082	
TL082CPWR	ACTIVE	TSSOP	PW	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082	Samples
TL082CPWRG4	ACTIVE	TSSOP	PW	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T082	Samples
TL082HIDDFR	ACTIVE	SOT-23-THIN	DDF	8	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	082F	Samples
TL082HIDR	ACTIVE	SOIC	D	8	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL082D	Samples
TL082HIPWR	ACTIVE	TSSOP	PW	8	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	082HPW	Samples
TL082ID	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL082I	
TL082IDG4	LIFEBUY	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL082I	
TL082IDR	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL082I	Samples
TL082IDRE4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL082I	Samples
TL082IDRG4	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL082I	Samples
TL082IP	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	TL082IP	Samples
TL082IPE4	ACTIVE	PDIP	P	8	50	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	TL082IP	Samples
TL082IPWR	ACTIVE	TSSOP	PW	8	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	Z082	Samples
TL082MFKB	ACTIVE	LCCC	FK	20	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962- 9851501Q2A TL082MFKB	Samples
TL082MJG	ACTIVE	CDIP	JG	8	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	TL082MJG	Samples
TL082MJGB	ACTIVE	CDIP	JG	8	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	9851501QPA TL082M	Samples
TL084ACD	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084AC	
TL084ACDE4	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084AC	
TL084ACDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084AC	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TL084ACDRE4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084AC	Samples
TL084ACDRG4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084AC	Samples
TL084ACN	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL084ACN	Samples
TL084ACNSR	ACTIVE	SO	NS	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084A	Samples
TL084BCD	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084BC	
TL084BCDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084BC	Samples
TL084BCDRG4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084BC	Samples
TL084BCN	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL084BCN	Samples
TL084BCNE4	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL084BCN	Samples
TL084CD	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084C	
TL084CDE4	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084C	
TL084CDG4	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084C	
TL084CDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084C	Samples
TL084CDRG4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084C	Samples
TL084CN	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL084CN	Samples
TL084CNE4	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	0 to 70	TL084CN	Samples
TL084CNSR	ACTIVE	SO	NS	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	TL084	Samples
TL084CPW	LIFEBUY	TSSOP	PW	14	90	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T084	
TL084CPWE4	LIFEBUY	TSSOP	PW	14	90	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T084	
TL084CPWR	ACTIVE	TSSOP	PW	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	0 to 70	T084	Samples
TL084HIDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084HID	Samples
TL084HIDYYR	ACTIVE	SOT-23-THIN	DYY	14	3000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	T084HDYY	Samples
TL084HIPWR	ACTIVE	TSSOP	PW	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084PW	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TL084ID	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL084I	
TL084IDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL084I	Samples
TL084IDRE4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL084I	Samples
TL084IDRG4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TL084I	Samples
TL084IN	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	TL084IN	Samples
TL084INE4	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	TL084IN	Samples
TL084MFK	ACTIVE	LCCC	FK	20	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	TL084MFK	Samples
TL084MFKB	ACTIVE	LCCC	FK	20	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962- 9851503Q2A TL084 MFKB	Samples
TL084MJ	ACTIVE	CDIP	J	14	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	TL084MJ	Samples
TL084MJB	ACTIVE	CDIP	J	14	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962-9851503QC A TL084MJB	Samples
TL084QD	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084Q	
TL084QDG4	LIFEBUY	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084Q	
TL084QDR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084Q	Samples
TL084QDRG4	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 125	TL084Q	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSELETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".